Name:___________

A2 Physics Unit 4

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /100	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	Max UMS
2017	31	40	50	60	70	90
2018	28	37	46	55	65	85
2019	27	36	45	54	64	84
2022	10	17	24	32	41	59 (/80)
2023	20	31	42	53	64	84

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Surname	Centre Number	Candidate Number
Other Names		2

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PHYSICS – A2 unit 4
Fields and Options

WEDNESDAY, 21 JUNE 2017 – MORNING

2 hours

	For Examiner's use only		
	Question	Maximum Mark	Mark Awarded
Section A	1.	20	
	2.	12	
	3.	17	
	4.	19	
	5.	12	
Section B	Option	20	
	Total	100	

1420U401 01

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator and a **Data Booklet.**

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Answer **all** questions.

Write your name, centre number and candidate number in the spaces at the top of this page.

Write your answers in the spaces provided in this booklet. If you run out of space, use the continuation page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

This paper is in 2 sections, **A** and **B.**

Section **A**: 80 marks. Answer **all** questions. You are advised to spend about 1 hour 35 minutes on this section.

Section **B**: 20 marks. Options. Answer **one option only.** You are advised to spend about 25 minutes on this section.

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **4*(b)*(ii)**.

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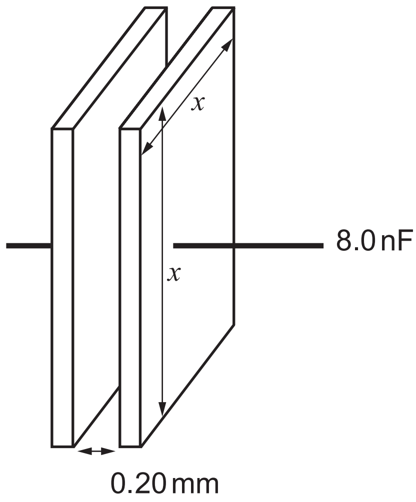
SECTION A

Answer **all** *questions.*

1. *(a)* (i) An 8.0 nF capacitor has square plates separated by 0.20 mm of air. Calculate the length (x) of the sides of the plates. [3]



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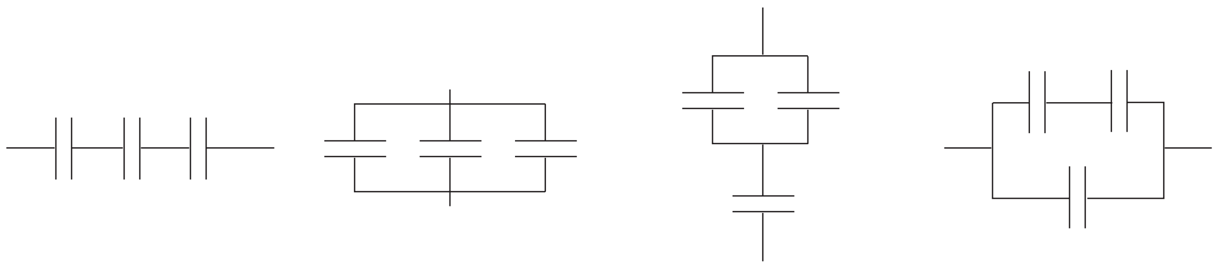
(ii) Which of the following combinations of 8.0 nF capacitors has a capacitance of 12.0 nF? Justify your answer. [4]

All individual capacitors shown are 8.0 nF.



1 2 3 4

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(iii) State how the capacitance of the capacitor can be increased without changing its dimensions. [1]

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(iv) Calculate the charge on the plates of the 8.0 nF capacitor when it stores an energy of 0.62 μJ. [3]

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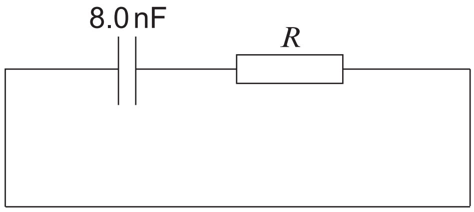
(b) (i) A charged 8.0 nF capacitor is discharged through a large resistor. Calculate the resistance of the resistor if the time constant is 15.5 ms. [2]



(ii) Calculate the fraction of the initial charge remaining on the capacitor after 31.0 ms. [3]

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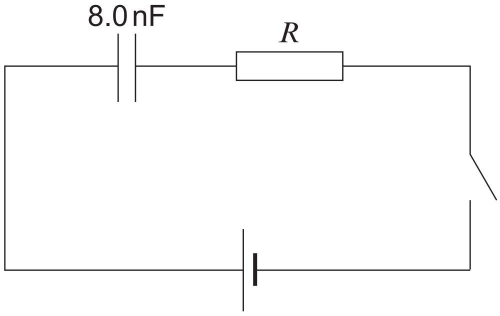
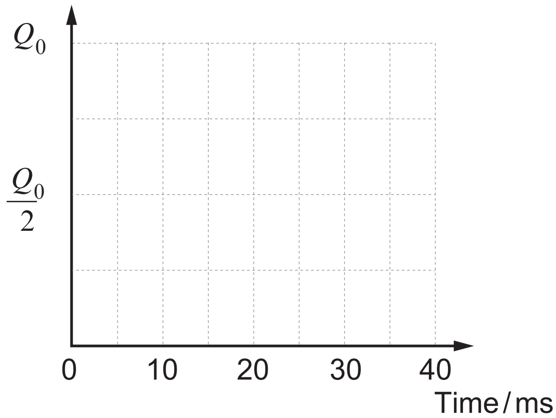
(c) An identical, uncharged 8.0 nF capacitor is **charged** in the following circuit, **using the same large resistor, R**. The switch is closed and the capacitor is uncharged at time $t = 0$.

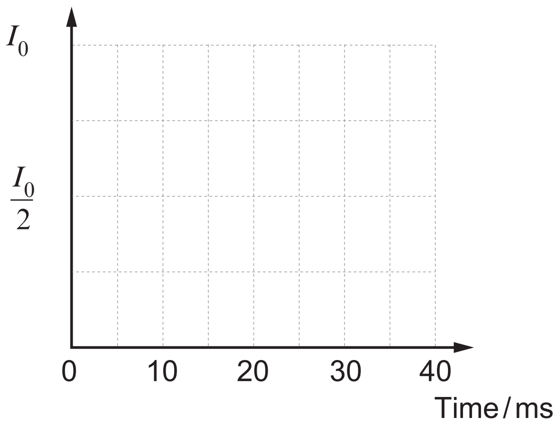


Without further calculations, sketch graphs of:

(i) the charge on the plates of the capacitor against time; [2]



(ii) the current in the circuit against time. [2]



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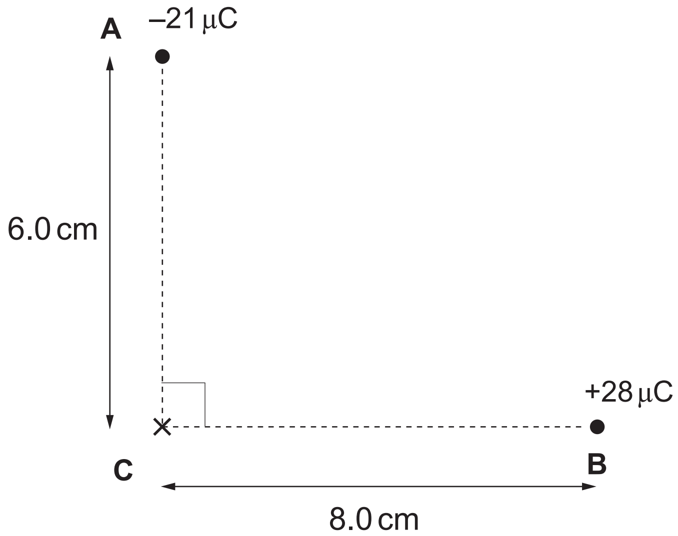
2. Two charges are placed at points **A** and **B** as shown in the diagram.



(a) Draw **two** arrows on the diagram to represent the directions of the electric fields at **C** due to the –21 μC and +28 μC charges. [2]

(b) Calculate the magnitude and direction of the resultant electric field at **C**. [5]

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(c) Show that the electrical potential at point **C** is zero. [2]

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(d) A positron is initially at rest at point **C** and accelerates rapidly due to the electric field. Calculate the potential at the point where the velocity of the positron is $5.4 \times 10^7\,\text{m}\,\text{s}^{-1}$. [3]

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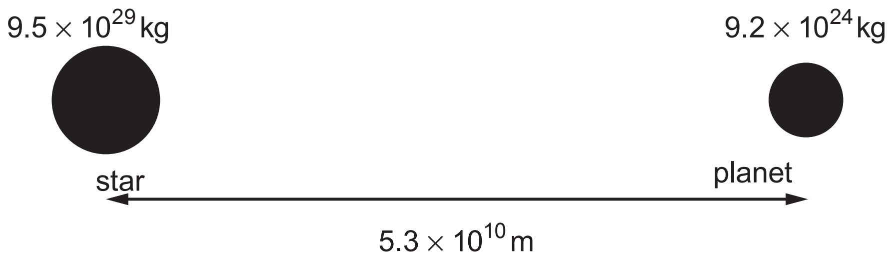
3. The planet (Kepler-186f) orbits the star (Kepler-186) and is considered to be the most Earth-like planet yet discovered. Approximate details of the system are shown in the diagram below.



(a) (i) Show that the radius of orbit of the star around the centre of mass of the system is 513 km. [2]

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(ii) The period of orbit of the planet is 130 days. Calculate the orbital velocity of the star. [3]

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(iii) The velocity calculated in part *(a)*(ii) is small. Explain why small velocities are difficult to measure using red shift. [2]

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(b) (i) Calculate the gravitational force between the star and the planet. [2]

(ii) An historic alternative to Newton's Law of Gravitation involved invisible steel rods connecting stars and planets and providing the force for keeping planets in orbit. A hypothetical cylindrical steel rod of radius 7.0×10^6 m and length 5.3×10^{10} m joins the planet to the star. Determine whether or not this steel rod is strong enough to keep the planet in orbit and explain why this theory is not accepted. (Breaking stress of steel = 4.5×10^8 Pa.) [4]

(c) (i) Calculate the potential energy of the star-planet system (see diagram in part *(a)*). [2]

(ii) In fact the planet has a slightly elliptical orbit. Explain how conservation of energy applies to this elliptical orbit. [2]

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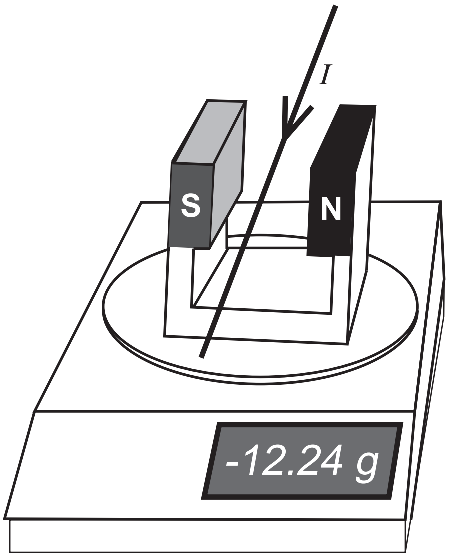
4. A student carries out an investigation. She uses a top pan balance to investigate the relationship $F=BI\ell$ using the set up shown in the diagram. The current-carrying wire is placed perpendicular to the magnetic field, B.

The current in the wire is varied using a variable power supply while the magnetic field, B, and the length of wire in the field, ℓ, are kept constant. Before switching on the current, the balance is set to zero.



(a) (i) Explain why the reading on the balance is negative for the direction of current and magnetic field shown in the diagram. [3]

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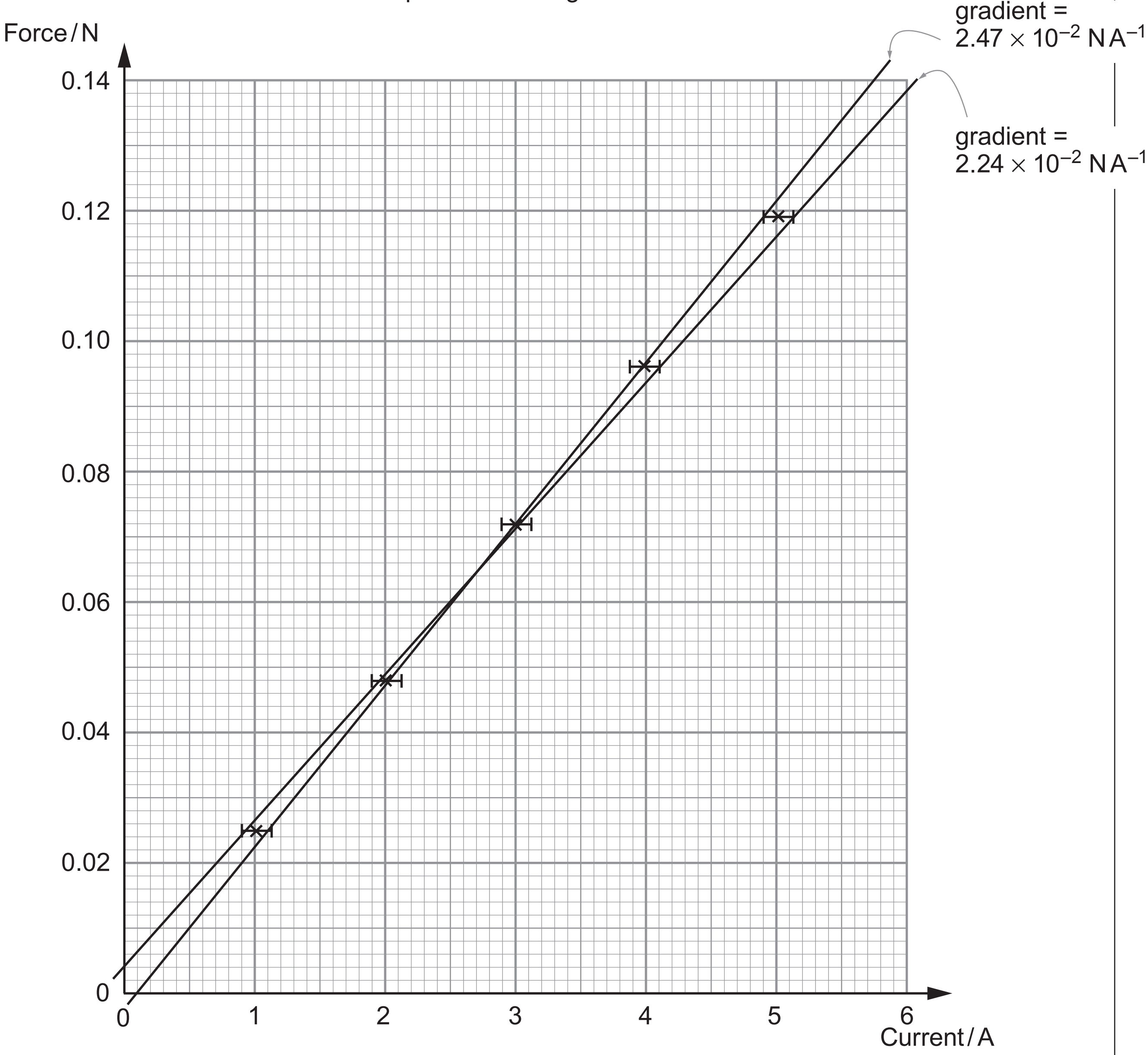
The student's results are plotted on the grid.



(ii) Calculate the mean value of the gradients (2.47×10^{-2} N A^{-1} and 2.24×10^{-2} N A^{-1}) and its **percentage** uncertainty. [3]

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(iii) The length of wire in the uniform magnetic field is (8.0 ± 0.5) cm. Determine a value for the magnetic field strength and its **absolute** uncertainty, stating both to an appropriate number of significant figures. [4]

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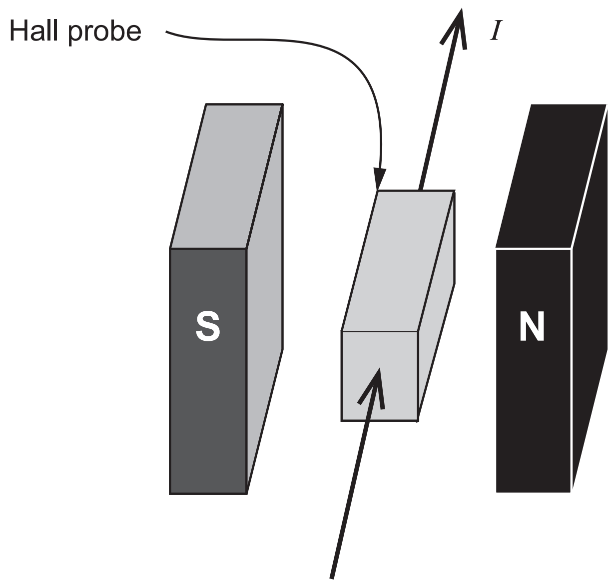
(b) In another experiment a Hall probe is used to confirm the value of the magnetic field strength obtained. The value of B obtained using the Hall probe is (310 ± 8) mT.



(i) Explain how the Hall voltage arises in the Hall probe shown. [3]

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(ii) Discuss the effectiveness of the experiment in part *(a)* for confirming the relationship $F=BI\ell$ and measuring the magnetic field strength, compared with using the Hall probe {see *(a)*(iii) and *(b)*}. [6 QER]

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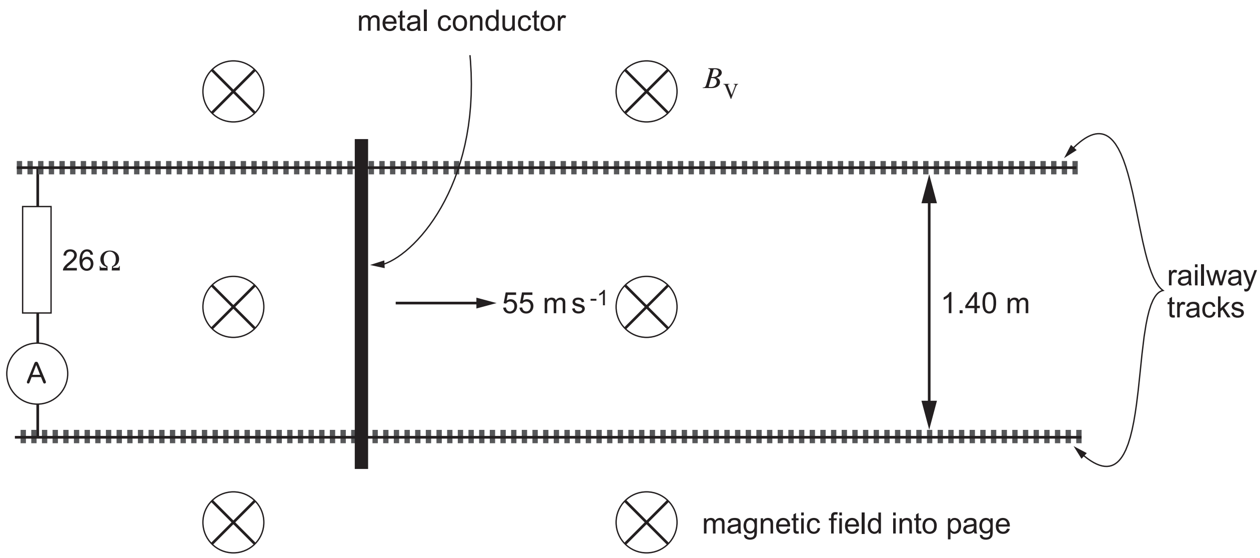
5. A metal conductor is placed across horizontal railway tracks and moved quickly in the direction shown. This set up is designed to measure the vertical component (B_V) of the Earth's magnetic field.



(a) (i) Explain why a current is detected by the ammeter. [2]

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(ii) State why the current is independent of the horizontal component of the Earth's magnetic field. [1]

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(b) Calculate a value for the vertical component of the Earth's magnetic field given that the reading on the ammeter is 184 μA. [3]

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(c) State the direction of the current in the resistor and how you obtained this direction. [2]

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(d) A student suggests that the opposing force due to the magnetic field on the moving conductor is negligible compared with other resistive forces. Is the student correct? Justify your answer with a calculation. [4]

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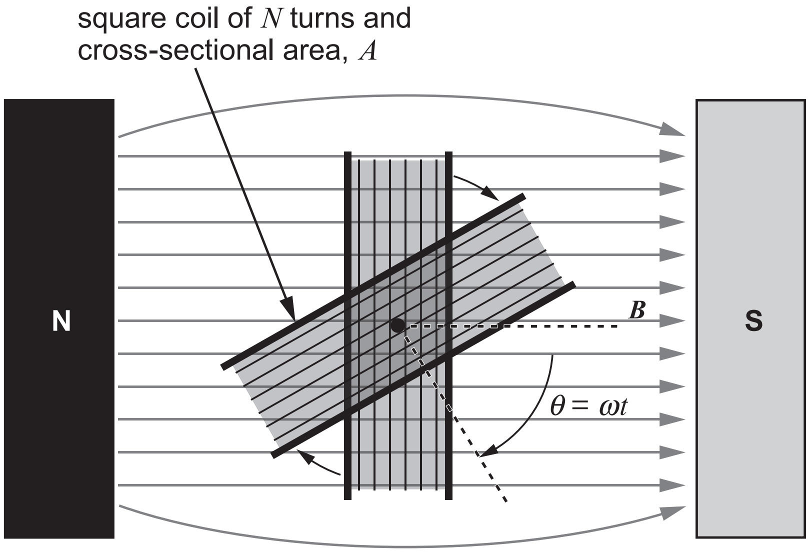
Option A – **Alternating Currents**

6. *(a)* A coil is rotated in a magnetic field.



(i) Use Faraday's law to explain why the emf is proportional to the angular velocity of the coil. [2]

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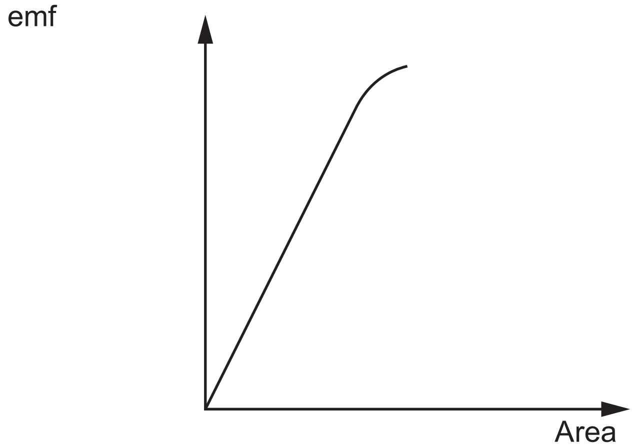
The peak emf induced in a rotating coil is given by:

$$E = \omega BAN$$

An experiment is carried out to investigate the variation in the peak emf with area, A, by using coils with different cross-sectional areas. The following graph is obtained.



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(ii) Discuss the agreement of the graph with the equation for the peak emf and suggest a reason for any disagreement (see diagram of coil opposite). [3]

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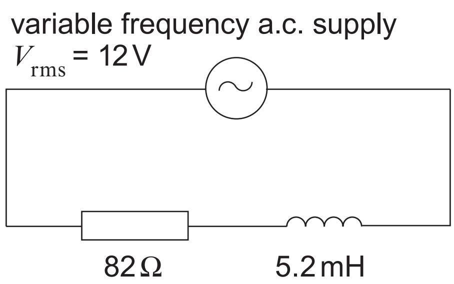
(b) (i) The minimum impedance of the circuit below occurs when the frequency of the a.c. supply is zero. Explain why this is so. [2]



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(ii) Calculate the rms current when the rms pd across the resistor and inductor are equal. [3]

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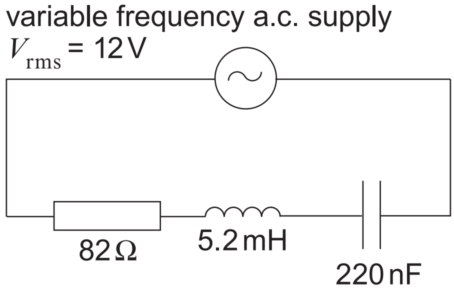
(c) (i) Explain how resonance arises in the LCR circuit shown. [3]



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(ii) Calculate the power dissipated in the circuit at resonance. [2]

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(iii) Calculate the rms current when the frequency of the supply is 9.40 kHz. [3]

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(iv) Discuss briefly the effectiveness of the given LCR circuit for providing a sharp resonance curve. [2]

(It may be useful to note that $\frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$.)

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Option B – **Medical Physics**

7. *(a)* An X–ray machine is designed to fire enormous numbers of electrons per second at very high speeds into a metal target. To do this it typically operates with a current of 100 mA. The electrons are accelerated from rest through a distance of about 3 cm hitting the target at half the speed of light.

(i) State how fast moving electrons can produce X-rays. [1]

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(ii) Determine the number of electrons arriving at the target every second. [2]

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(iii) Calculate the acceleration of an electron in the X-ray tube as it travels towards the target. [3]

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(iv) Explain how the photon energy and the intensity of the X-ray output can be increased. [2]

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(b) Give **one** advantage and **one** disadvantage of using CT scans compared with conventional X-ray imaging. [2]

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(c) The following table provides information relating to ultrasound in fat and muscle.

Tissue	Density / $kg\,m^{-3}$	Speed / $m\,s^{-1}$
Muscle	1.1×10^3	1.6×10^3
Fat	0.8×10^3	1.5×10^3

By considering acoustic impedance, explain whether or not you would expect to receive an ultrasound echo from a muscle-fat boundary. [4]

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(d) (i) Explain briefly the role played by the absorbed radio waves in the resonance process of MRI scans. [2]

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(ii) In an MRI scanner radio waves of frequency 64 MHz are absorbed and cause hydrogen atoms to resonate in a magnetic field of 1.5 T. Determine the frequency of absorbed radio waves that would be needed if the field was reduced to 1.2 T. [2]

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(e) (i) Alpha particles have a radiation weighting factor of 20. If a patient receives an absorbed dose of 55 mGy determine the dose equivalent received. [1]

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(ii) State whether or not beta particles would have a weighting factor of 20, greater than 20 or less than 20. Justify your answer. [1]

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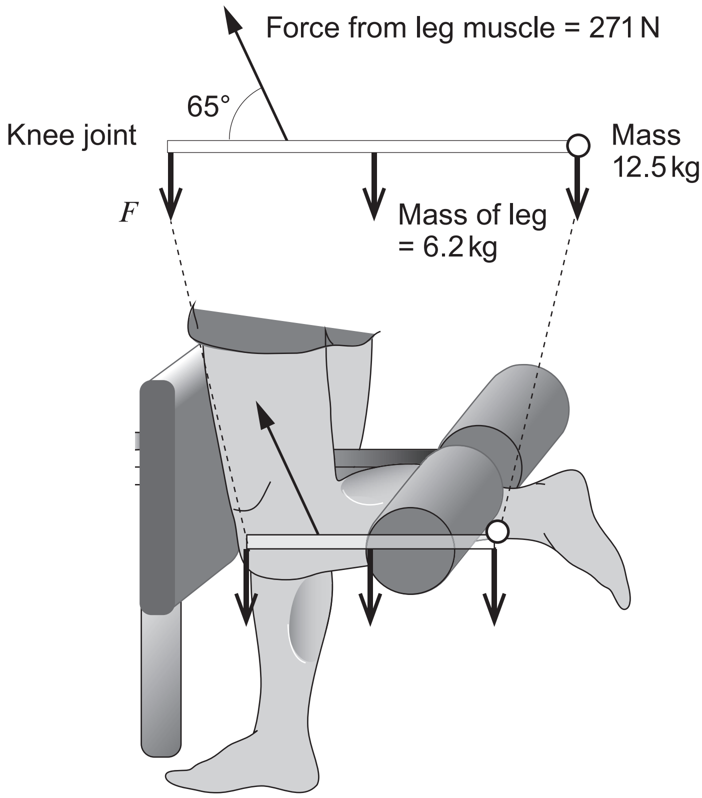
Option C – **The Physics of Sports**

8. The following simplified diagram is of the lower leg of an athlete holding a mass of 12.5 kg in equilibrium. The leg pivots around the knee joint and the effective mass of the lower leg is 6.2 kg. The force, F, shown is the vertical component of the force exerted on the lower leg by the knee.



(a) Calculate the force, F. [3]

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(b) (i) A football player must kick a football a distance of 43.6 m for it to be a successful kick. Determine whether or not the kick will be successful if it is kicked with a speed of 18 m s^{-1} at an angle of 50° to the horizontal. *Ignore the effects of air resistance.* [4]

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(ii) The football has a mass of 0.45 kg and spins at a rate of 8.8 Hz. It has a moment of inertia of 0.0041 kg m^2.

I. Explain what is meant by the moment of inertia. [2]

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II. Calculate the angular momentum of the football. [2]

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III. Determine the **total** kinetic energy of the ball **at its greatest height** *(remember to ignore the effects of air resistance)*. [4]

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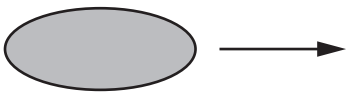
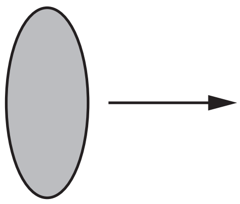
(c) Due to the shape of a rugby ball, its cross-sectional area can be varied depending on its orientation and motion. The minimum and maximum cross-sectional areas are 43.2 cm^2 and 217.7 cm^2 respectively and are shown below.



Minimum cross-sectional area = 43.2 cm^2



Maximum cross-sectional area = 217.7 cm^2

(i) Explain clearly by what factor the drag force changes for these two cross-sectional areas. [3]

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(ii) If the rugby ball had been kicked in a region where the density of the air is lower, explain how your answer to part *(c)*(i) would change, if at all. [2]

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Option D – **Energy and the Environment**

9. *(a)* (i) The total power emitted by the Sun is 4.0×10^{26} W. Given that the Earth is 1.5×10^{11} m from the Sun show that the intensity of the radiation reaching the Earth's outer atmosphere is approximately $1400\,\mathrm{W\,m^{-2}}$. [2]

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(ii) The mean annual power available at the Earth's surface in the UK is only about 20% of the answer to part *(a)*(i). By the middle of 2015 solar photovoltaic (PV) installations accounted for approximately 8 GW of the UK's electricity production. Given that solar panels are only 15% efficient at converting solar radiation into electricity, estimate the total area of solar panels in the UK in 2015. [2]

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(iii) Give a reason why the amount of electrical energy produced by a solar farm in Portugal is likely to be greater than that from a solar farm of the same size in the UK per year. [1]

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(b) (i) State Wien's law. [1]

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(ii) Assuming a mean temperature of 290 K for the Earth and that it emits radiation from its surface as a black body, show that the peak wavelength of the radiation emitted by the Earth is in the infra-red region of the electromagnetic spectrum. [2]

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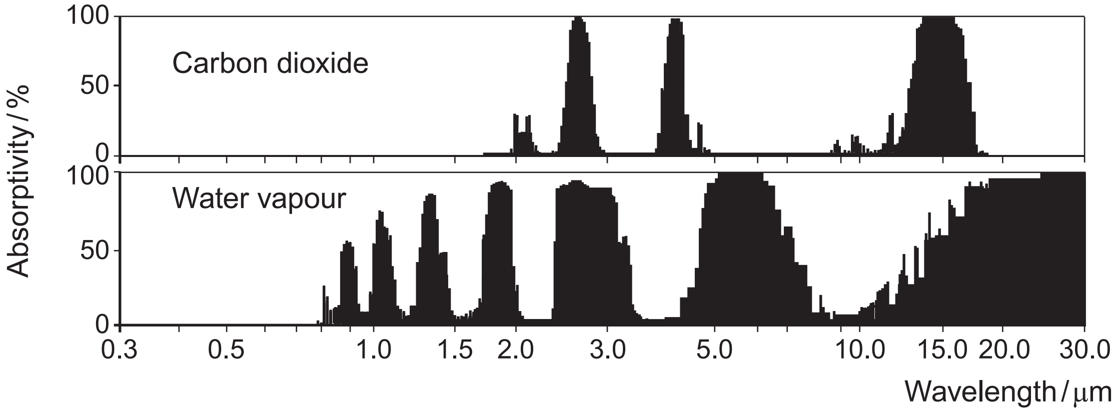
(iii) The graphs below show greenhouse gas absorption spectra for water vapour and carbon dioxide as a function of the wavelength of the radiation incident on the gas. An absorptivity of zero % means no radiation is absorbed, whilst an absorptivity of 100% means that all the incident radiation is absorbed.



www.worldclimatereport.com

I. Use the graphs to explain how increased levels of water vapour and carbon dioxide in the atmosphere can account for the greenhouse effect. [4]

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II. The term *'positive feedback'* is used to describe a system's output influencing the system so as to increase the output even further. The Earth's surface and atmosphere may be regarded as a system, and temperature as the system's output. Explain how the greenhouse effect is part of a positive feedback. [2]

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(c) (i) Define the term U-value and state **one** factor which affects the U-value of a material. [2]

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(ii) A room in a large office block is maintained at a constant temperature of 21 °C. The room has one of its walls, measuring 5.0 m × 3.0 m, exposed to the outside. This wall is double glazed (made of two layers of glass with a small air filled gap between them). Negligible heat is lost through the other walls of the room.
Calculate the energy per second transmitted through the outside wall if the outside temperature is 9 °C. [U-value of double glazing glass = 1.8 W m^{-2} K^{-1}] [2]

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(iii) The heating system in the room can provide a maximum output of 1 kW. Determine the lowest outside temperature for which the double glazing can maintain an internal temperature of 21 °C. [2]

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END OF PAPER

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Surname	Centre Number	Candidate Number
Other Names		2

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PHYSICS – A2 unit 4
Fields and Options

FRIDAY, 8 JUNE 2018 – MORNING

2 hours

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	Question	Maximum Mark	Mark Awarded
Section A	1.	18	
	2.	18	
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	5.	15	
Section B	Option	20	
	Total	100	

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Section **B**: 20 marks. Options. Answer **one option only.** You are advised to spend about 25 minutes on this section.

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **3*(d)***.

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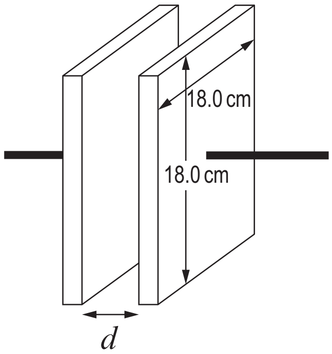
SECTION A

Answer **all** *questions.*

1. *(a)* (i) A laboratory technician has two square aluminium plates with sides of length 18.0 cm. Calculate the separation, d, of the plates that produces a capacitance of 2.0 nF. [3]



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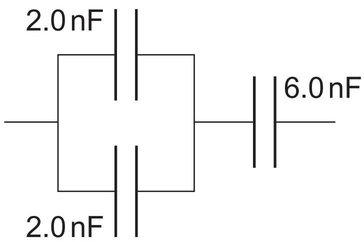
(ii) Calculate the capacitance of the capacitor combination shown. [3]



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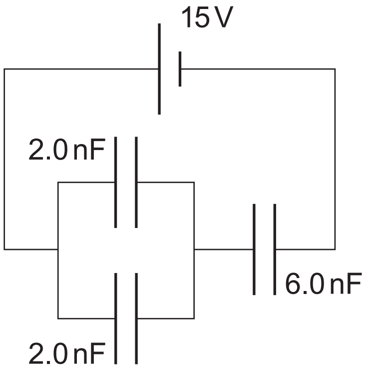
(iii) Show that the charge stored on a 2.0 nF capacitor in the following circuit is 18 nC. [3]



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(iv) Calculate the total energy stored by the capacitor combination. [2]

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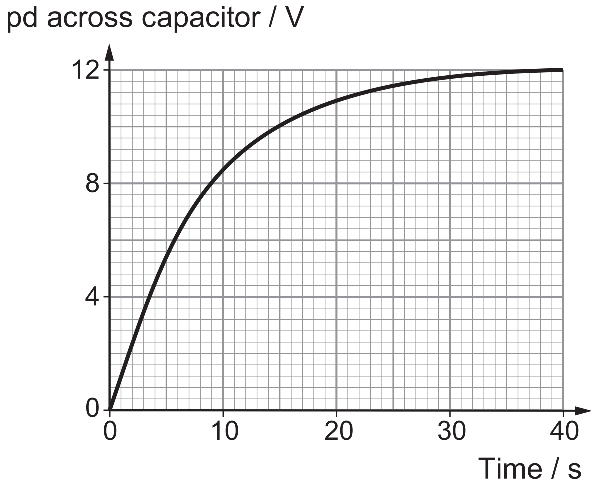
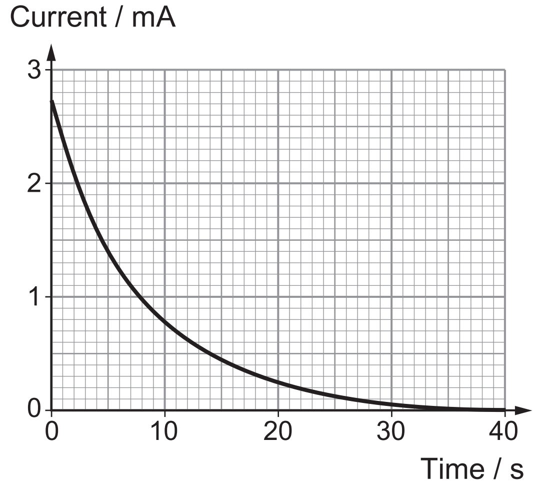
(b) A group of students investigated the **charging** of a capacitor through a resistor and obtained the following data.





(i) Draw a circuit diagram of the apparatus that might have been used to obtain the data. [2]

(ii) Use the graphs to determine:

- the emf of the power supply;
- the resistance of the resistor;
- the capacitance of the capacitor. [5]

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2. *(a)* The escape velocity, v, of a mass, m, from a spherical mass, M, and radius, R, can be calculated using:

$$\tfrac{1}{2}mv^2 - \frac{GMm}{R} = 0$$

(i) Explain how this equation is an application of conservation of energy. [3]

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(ii) Calculate the escape velocity from the Sun ($M_{\text{Sun}} = 1.99 \times 10^{30}$ kg, $R_{\text{Sun}} = 6.96 \times 10^{8}$ m). [3]

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(b) (i) The temperature of the surface of the Sun is 5 780 K. Use a kinetic theory equation to show that the rms speed of a free electron on the surface of the Sun is approximately 500 km s^{-1}. [4]

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(ii) By considering your answers to *(a)*(ii) and *(b)*(i), explain why the Sun has a slight positive charge. [2]

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(iii) A student claims that a positive charge of approximately 0.08 C on the Sun is enough to produce an electrostatic force equal to the gravitational force on an escaping electron. Determine whether or not she is correct. [3]

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(iv) Estimate the percentage of lost electrons compared with the total number of electrons on the Sun. Assume that the Sun is mainly hydrogen and that it has lost 0.08 C of charge in the form of electrons. [3]

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3. *(a)* Explain why the age of the Universe can be approximated as $\frac{1}{H_0}$, where H_0 is the Hubble constant. [2]

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(b) The megaparsec (Mpc) is a unit of astronomical distance equal to 3.09×10^{22} m. Use Hubble's law to show that the expected redshift for a supernova at a distance of 1 Mpc for a wavelength of 486.1 nm is approximately 0.11 nm. [3]

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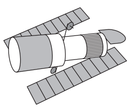
(c) The spiral galaxy shown is rotating anticlockwise and is viewed by the Hubble Space Telescope.

Hubble Space Telescope



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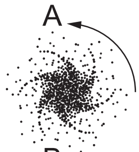

Rotating spiral galaxy

The measured **blue** shift at point A of the galaxy is 0.22 nm (for the 486.1 nm wavelength) and the measured **redshift** at point B of the galaxy is 0.66 nm (for the same wavelength). Calculate the recessional velocity of the galaxy and the rotational speed of the galaxy at A (assume that A and B have the same rotational speeds). [5]

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(d) Explain how spiral galaxies provide evidence for the existence of dark matter and how this evidence has been gathered. [6 QER]

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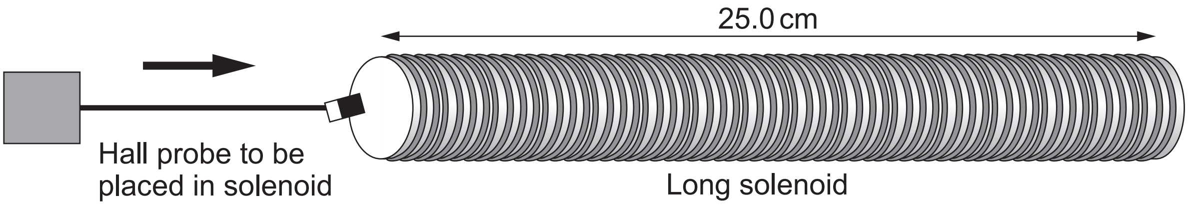
4. An experiment is carried out to measure the magnetic field in a long solenoid as the current in the solenoid is varied.



Theory states that the relationship between magnetic flux density, B, at the centre of the solenoid and current, I, is given by the equation:

$$B = \mu_0 n I$$

The results obtained are shown in the table and plotted on the grid opposite along with error bars and lines of maximum and minimum gradient.

Current / A ±0.01 A	Magnetic flux density / mT ±2 mT
0.50	11
1.00	20
1.50	29
2.00	41
2.50	49

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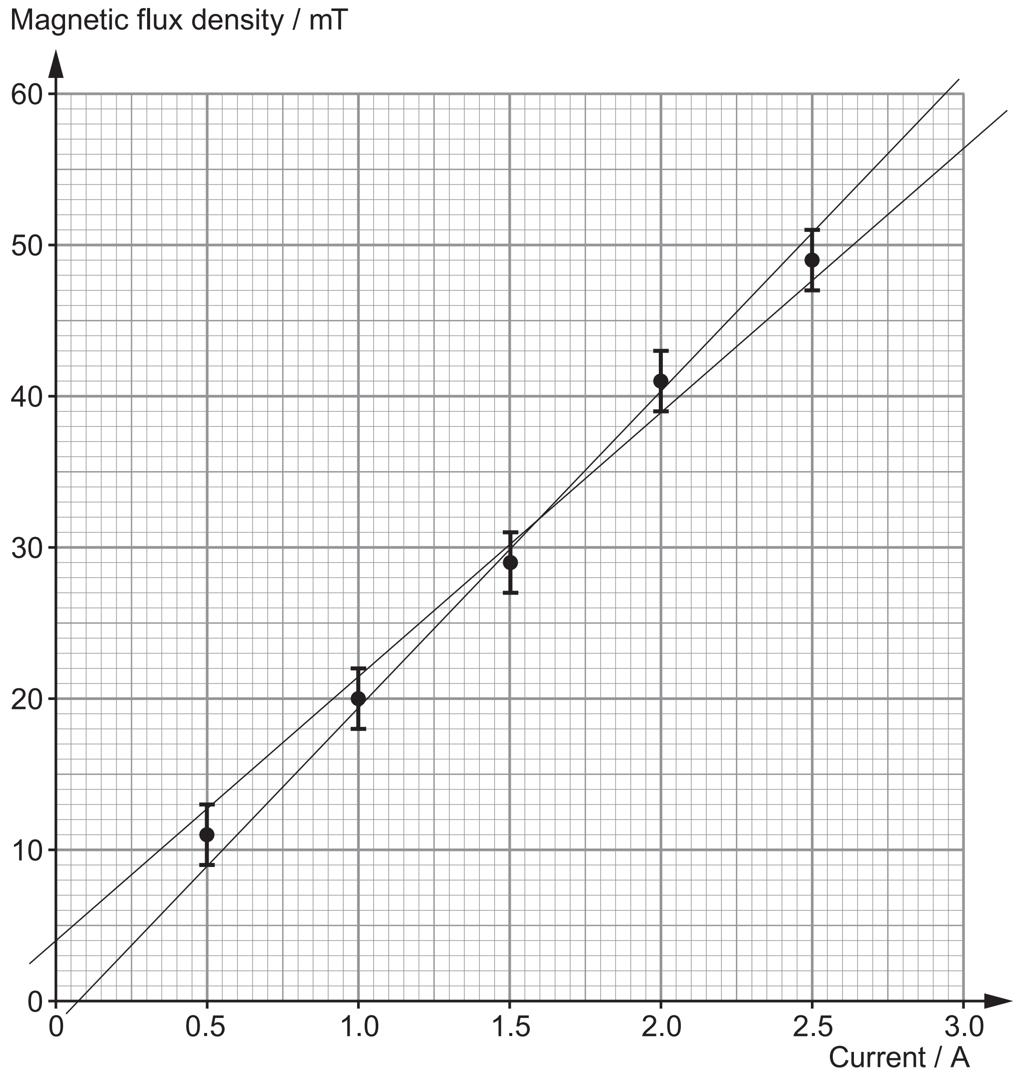

(a) Calculate the mean value of the gradient along with its **percentage** uncertainty. [4]

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(b) Calculate the number of turns in the solenoid along with its **absolute** uncertainty. The percentage uncertainty in the length of the solenoid is 0.4%. [3]

(c) (i) The solenoid manufacturer states that there are exactly 5000 turns in the solenoid. Evaluate the accuracy of your value obtained in part *(b)* and whether or not the graph is in agreement with the equation: $B = \mu_0 nI$. [4]

(ii) Suggest a reason for the disagreement between the manufacturer's stated value (5000 turns) and your value calculated in part *(b)*. Suggest how the experimental technique might be improved for better agreement. [2]

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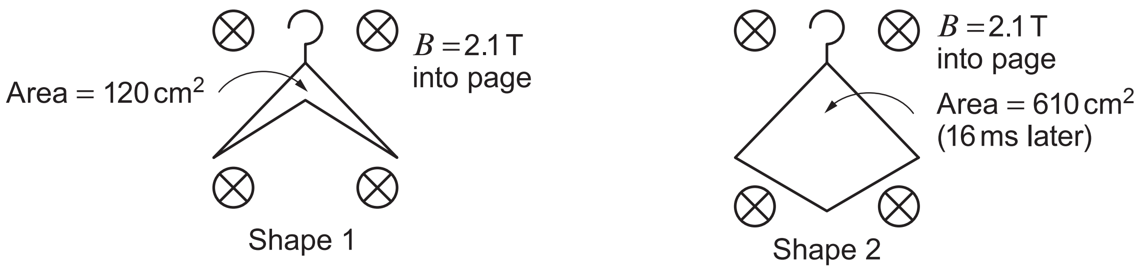
5. An experiment is carried out in a very strong uniform magnetic field in order to confirm Faraday's Law under extreme conditions. A coat hanger made of aluminium wire is bent from Shape 1 to Shape 2 in a time of 16 ms.



(a) (i) Show that a mean emf of 6.4 V is induced in the coat hanger. [2]

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(ii) Show on the diagram of Shape 2 the direction of the induced current and state very briefly how you determined this direction. [2]

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(b) The aluminium wire has a circular cross-section of diameter 3.0 mm and the length of the wire through which the current flows is 91 cm. Show that the mean current in the wire is approximately 1 900 A (resistivity of aluminium = $2.65 \times 10^{-8}\ \Omega$ m). [3]

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(c) Iestyn claims that changing the shape of the coat hanger from Shape 1 to Shape 2 in 16 ms in the magnetic field will increase its temperature by less than 1 °C. Determine, using appropriate calculations, whether or not Iestyn is correct. (Density of aluminium = 2 700 kg m^{-3}, specific heat capacity of aluminium = 897 J kg^{-1} K^{-1}.) [5]

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(d) For medical research, it is decided to investigate the effect of this strong magnetic field (2.1 T) on patients with metal replacement joints to see if the metal joints become hot or undergo large forces (during MRI scans). Discuss the ethics of such an experiment. [3]

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Option A – **Alternating Currents**

6. *(a)* The sinusoidal emf produced by a rotating coil in a magnetic field has a **peak** emf of 18.0 V and a frequency of 70.0 Hz. The coil is connected to the y-input of an oscilloscope.

(i) The output of the rotating coil is used to power a small lamp of known resistance. Explain how you would calculate the rms input power to the lamp from the known resistance and **peak** emf. [2]

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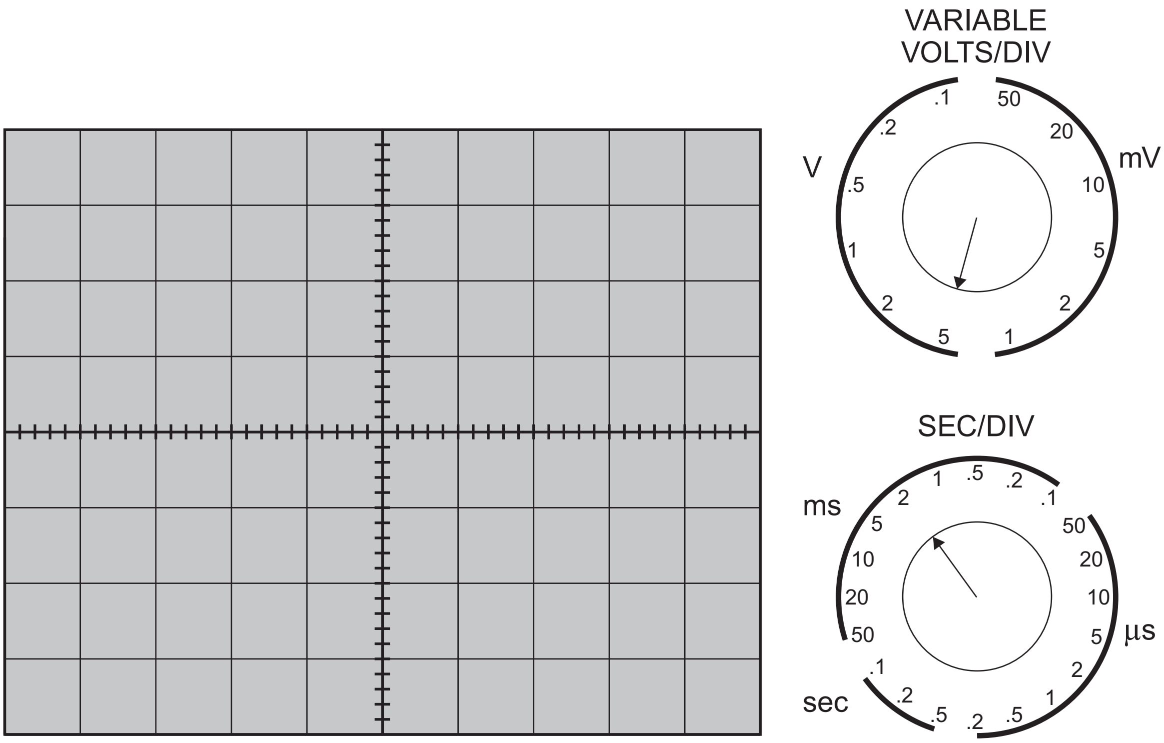
(ii) Draw a typical trace that might be seen on the oscilloscope screen with the oscilloscope settings shown. [5]

Space for calculations.



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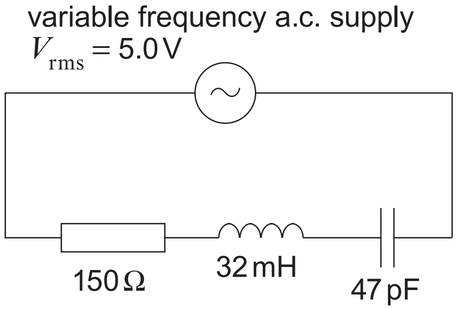
(b) An LCR circuit is shown below.



(i) Explain why the resonance frequency of the circuit occurs when: [3]

$$\omega L = \frac{1}{\omega C}$$

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(ii) Calculate the resonance frequency of the circuit. [2]

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(iii) Calculate the rms current when the frequency of the supply is 324 kHz. [3]

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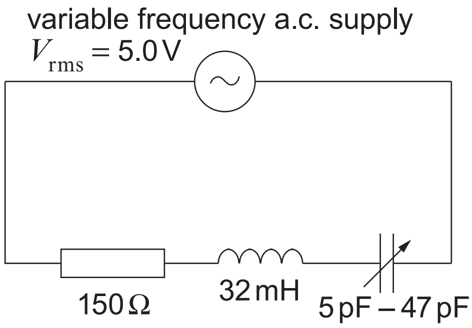
(iv) A student claims that the following circuit cannot have a peak pd above 1.5 kV across the capacitor. Investigate whether or not he is correct. [5]

(It may be useful to note that $\frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$.)



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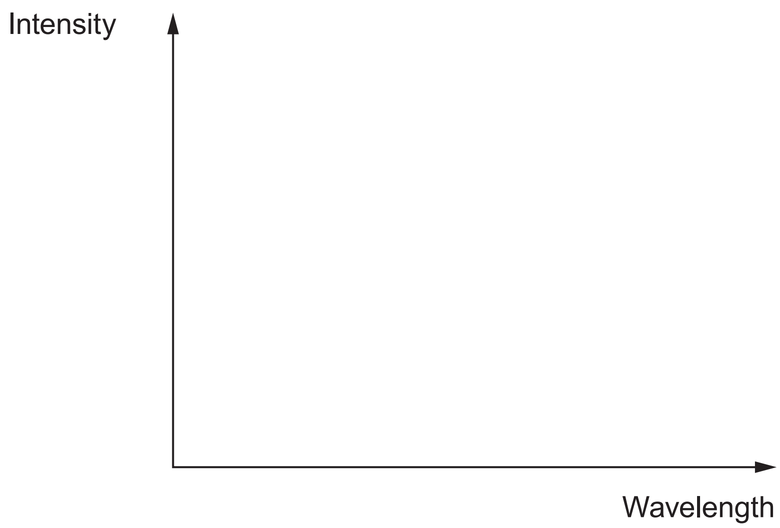
Option B – **Medical Physics**

7. An X-ray machine has a working potential difference of 75 000 V.

(a) (i) Sketch a graph of intensity against wavelength for the resulting X-ray spectrum. Label the main features of this spectrum, including a value for the minimum wavelength. [2]

Space for calculation



(ii) At the working potential difference the current in the tube is 120 mA and the efficiency of the X-ray machine is 0.7 %. Calculate the rate of production of heat. [2]

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(iii) A metal plate of thickness 1.4 mm is used to reduce the intensity of the X-rays produced to 60 % of the incident intensity. If a second identical plate is now also placed in the beam, calculate the new transmitted **percentage** intensity. [3]

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(b) (i) Describe how the Doppler shift principle can be used to measure the speed of blood through an artery. [2]

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(ii) Ultrasound of frequency 2 MHz was used to calculate the speed of blood and a Doppler shift of 200 Hz was detected. The measurement was taken at an angle of 40^{o} to the direction of flow and the speed of ultrasound through the blood is $1500\,m\,s^{-1}$. Calculate the speed of blood flow. [2]

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(c) (i) Describe the properties of technetium–99m (Tc–99m) that make it such a good radioisotope in the effective diagnosis of medical problems. Justify your choice of properties. [3]

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(ii) Explain clearly how a gamma camera is used to detect the gamma rays given off by a technetium–99m source. [3]

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(iii) In positron emission tomography a positron annihilates an electron producing two photons of energy 0.511 MeV. By setting out your reasoning clearly, determine whether or not the value of the photon energy is correct. [3]

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Option C – **The Physics of Sports**

8. *(a)* Define angular acceleration. [2]

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(b) A gymnast begins a routine to dismount from the horizontal bar by increasing her angular velocity from $3.4\,\text{rad}\,\text{s}^{-1}$ to $8.0\,\text{rad}\,\text{s}^{-1}$ in a time of 2.3 s. The moment of inertia of the gymnast is $34\,\text{kg}\,\text{m}^2$.

(i) Calculate the torque on the gymnast. [4]

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(ii) The gymnast lets go of the bar and somersaults in the air before she lands on her feet on the ground. Explain why the gymnast pulls in her arms and adopts a tuck position as she is in the air. [4]

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(iii) The gymnast, of mass 52 kg, lands on the mat with a velocity of $5.7\,\mathrm{m\,s^{-1}}$ and comes to rest in a time of 0.26 s. Show clearly that the mean force exerted by the gymnast on the mat is approximately 1 000 N. [3]

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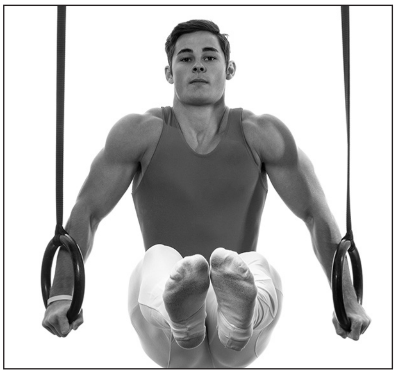
(c) In a separate event, a male gymnast performs a routine on the rings as shown.



(i) Explain in terms of centre of gravity why he does not rotate. [2]

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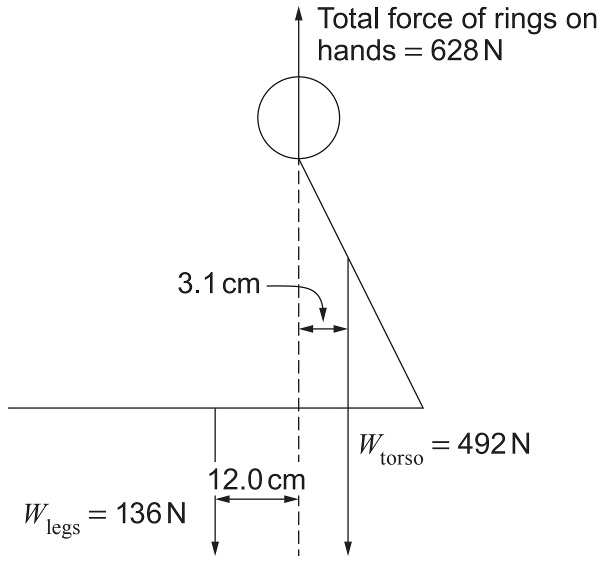
(ii) At a later point the forces acting on the gymnast are shown in the following diagram. Justify, using calculations, the motion of the gymnast. [5]



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Option D – **Energy and the Environment**

9. *(a)* (i) The Sun has a surface temperature of 5800 K and a radius of 7.0×10^8 m. Stating the name given to the law you use, show that the power radiated by the Sun (Solar Luminosity) is approximately 4×10^{26} W. [3]

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(ii) The main energy production mechanism in the sun is the proton-proton cycle. This consists of several fusion reactions, the net effect of which is to combine a number of protons to form one helium nucleus as shown:

$$\ldots\ldots\ldots\ldots \; {}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\nu_e + 2\,{}^{0}_{+1}\mathrm{e}$$

I. **Complete** the equation. [1]

II. Name the particle which has the symbol ${}^{0}_{+1}\mathrm{e}$. [1]

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(iii) The energy released in the reaction is 26.7 MeV. Use this information and the answer to *(a)*(i) to determine the mean rate of production of helium nuclei in the Sun. [2]

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(b) Due to absorption in the atmosphere, the maximum intensity of the Sun's radiation received at the Earth's surface in the UK is about 750 W m^{-2}. Show that this corresponds to approximately 50 % of the solar intensity reaching the Earth's atmosphere. [Sun-Earth distance = 1.50×10^{11} m]. [2]

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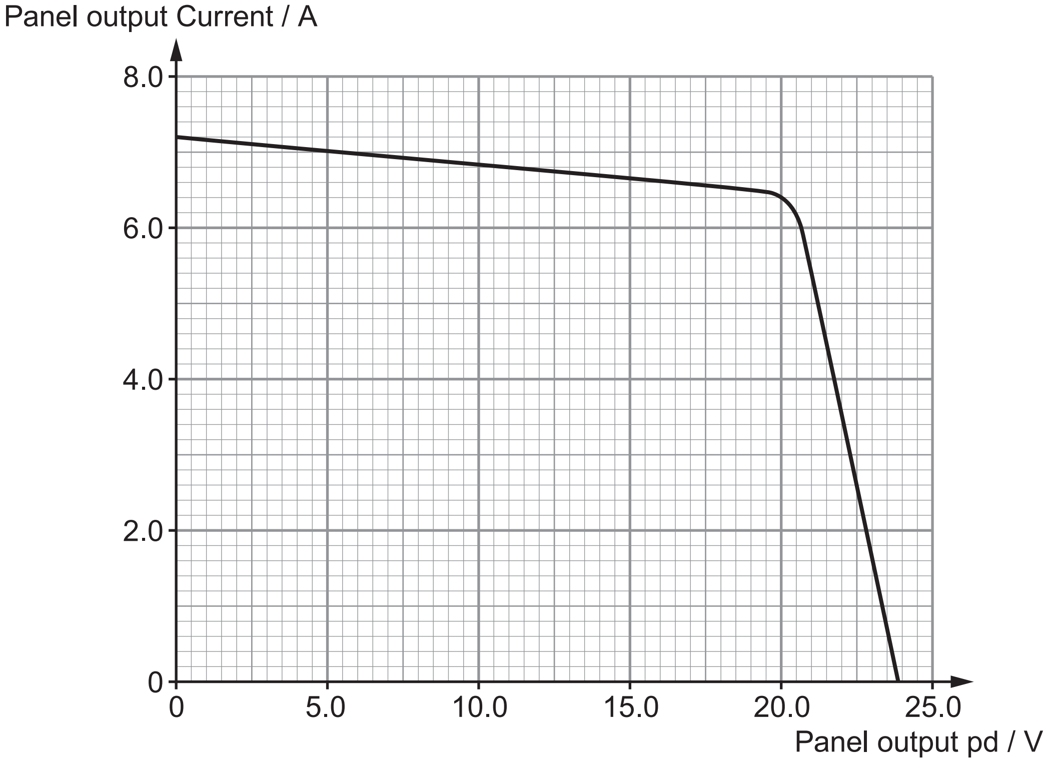
(c) Solar (PV) panels are used to produce electricity from the solar radiation incident upon them. The output power of PV panels depends on the load resistance and the intensity of the radiation. The graph shows the output characteristics for a solar panel of area 1 m^2 for varying values of load resistance for a constant **light power of 750 W**.



(i) Engineers designing this panel require that it produces at least 15 % of the maximum input power. Determine whether or not the panel meets this requirement when operating at maximum output power. [3]

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(ii) Determine the number of panels of this type needed to power a 1 kW electric kettle, and explain why, **in reality**, the actual number of panels needed will be greater. [3]

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(d) Scientists attempting to generate electricity by nuclear fusion on Earth must overcome a number of difficulties. One condition which needs to be satisfied is to ensure a high enough temperature.

(i) Explain in terms of energy and the interaction of particles why a high temperature is necessary. [3]

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(ii) For a particular nuclear fusion reaction to be successful the value of its *triple product* must be $\geq 2.6 \times 10^{28}\,\text{s K m}^{-3}$. Plasma of volume $75\,\text{m}^3$ contains 2.2×10^{22} reacting particles at a temperature of $120 \times 10^6\,\text{K}$. If a confinement time of 0.8 seconds is achieved, determine whether or not fusion is possible under these conditions. [2]

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END OF PAPER

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Surname	Centre Number	Candidate Number
Other Names		2

GCE A LEVEL

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PHYSICS – A2 unit 4
Fields and Options

MONDAY, 17 JUNE 2019 – MORNING

2 hours

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1.**	**27**	
	2.	**11**	
	3.	**12**	
	4.	**18**	
	5.	**12**	
Section B	**Option**	**20**	
	Total	**100**	

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ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Answer **all** questions.

Write your name, centre number and candidate number in the spaces at the top of this page.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

This paper is in 2 sections, **A** and **B**.

Section **A**: 80 marks. Answer **all** questions. You are advised to spend about 1 hour 35 minutes on this section.

Section **B**: 20 marks. Options. Answer **one option only**. You are advised to spend about 25 minutes on this section.

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **5*(b)***.

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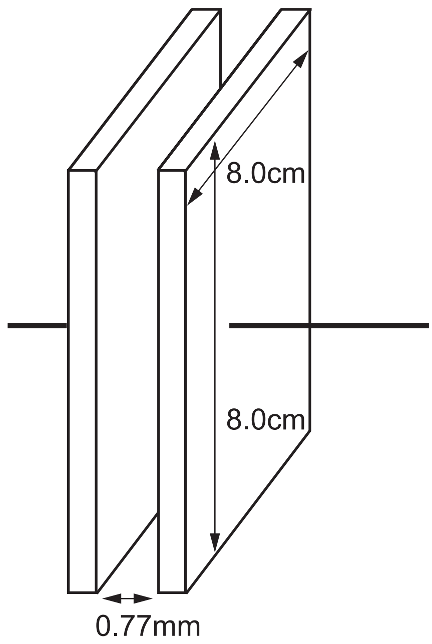
SECTION A

Answer **all** *questions.*

1. *(a)* (i) For the air-spaced parallel plate capacitor shown, calculate the pd applied when it stores 25.5 μJ of energy. [4]



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(ii) Explain why the capacitor stores energy when a pd is applied to the plates. [2]

(iii) A group of scientists claim that they have developed a new dielectric that enables the above capacitor to store a million times more charge and energy for a given pd. Explain what further steps must be taken by the scientific community and industry before this new dielectric can be used in devices and sold to the public. [3]

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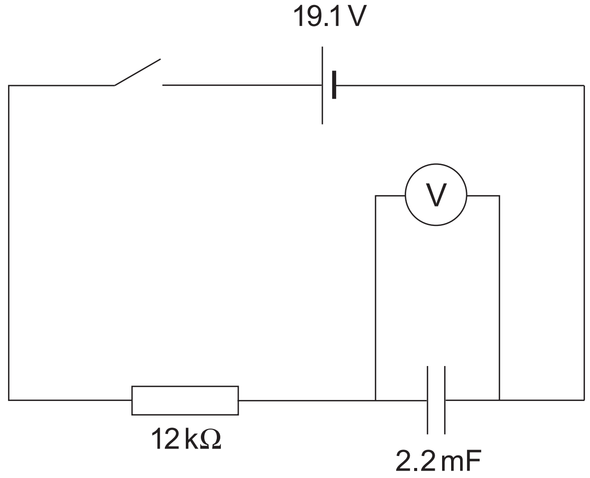
(*b*) Bethan investigates the charging of a capacitor using the following circuit.



The results she obtains are then tabulated.

Time / s	pd across capacitor / V ± 0.5 V	Charge on capacitor / mC ± 1.1 mC
10.0	6.1	13.4
20.0	10.0	22.0
30.0	12.8	....
40.0	15.1	33.2
50.0	16.2	35.6
60.0	17.1	....
70.0	17.7	38.9
80.0	18.3	40.3

(i) Confirm that the uncertainty in the charge is 1.1 mC (you may assume that the uncertainty in the 2.2 mF capacitor is negligible). [1]

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(ii) **Complete** the table. [2]

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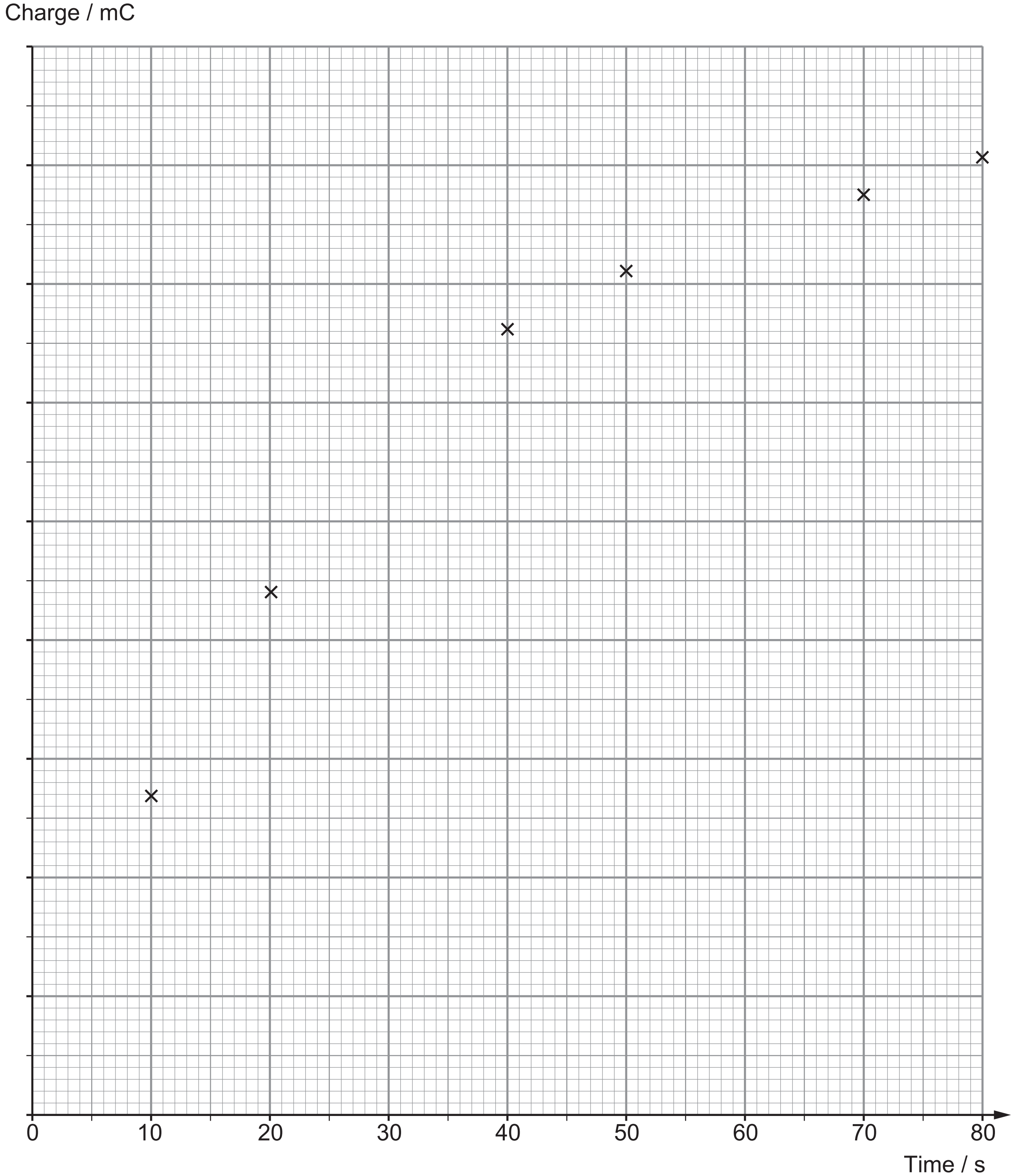
(iii) Complete the graph by labelling the y-axis scale, plotting the remaining two points, adding error bars for charge and drawing a curve of best fit. [5]



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(iv) **Use the curve of best fit** to obtain the time constant of the charging circuit (show your working and do not simply multiply the resistance by the capacitance). [3]

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(v) By drawing a suitable tangent, calculate the current in the circuit at 45 s. [2]

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(c) Use the graph on page 5 and your answers to *(b)*(iv) and *(b)*(v) to evaluate whether or not the data obtained in this experiment are in good agreement with the equations: [5]

$$Q = Q_0\left(1 - e^{-\frac{t}{RC}}\right) \textbf{ and } I = I_0 e^{-\frac{t}{RC}}$$

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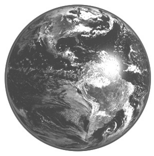
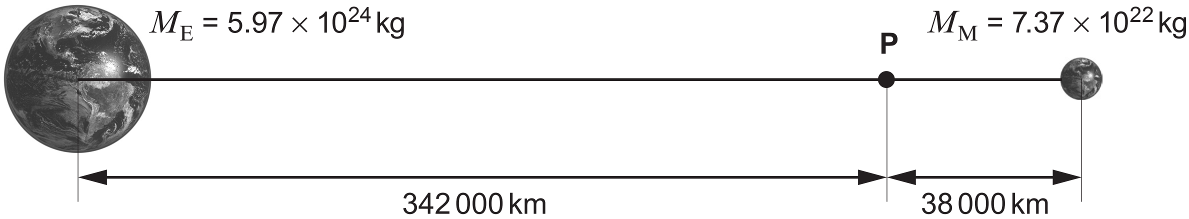
2. *(a)* Assuming that the Earth is an isolated perfect sphere, draw its gravitational field lines and equipotential surfaces. [3]



(b) (i) Use the information in the diagram to calculate the gravitational potential at point **P**. [3]



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(ii) Use the information in the diagram to show that the resultant gravitational field at point **P** is very small. [2]

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(iii) Myfanwy correctly calculates that the force on a 25 tonne spaceship would be negligible at point **P** and that the force would increase by approximately 0.5 N for every 10 km moved away from point **P** towards the Earth. Dafydd then concludes that the spaceship will perform simple harmonic motion about point **P**. Deduce whether or not Dafydd is correct **(no further calculations are required)**. [3]

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3. *(a)* (i) State Kepler's 1st and 2nd laws. [3]

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(ii) Kepler's 3rd law can be derived from Newton's gravitational law and the equation for centripetal motion. Show that, for any object in a circular orbit about the Earth:

$$T^2 = \frac{4\pi^2}{GM_E} r^3$$

where T = the period of orbit, r = the radius of orbit, G = the gravitational constant and M_E = the mass of the Earth. [3]

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(b) The radius of the geostationary orbit above the Earth's equator is 42 000 km and the Earth-Moon distance is 380 000 km. Use Kepler's 3rd law to calculate the period of the Moon's orbit. [3]

(c) Estimate the minimum possible orbital period for a satellite orbiting the Earth, stating any assumption that you make (the radius of the Earth is 6 370 km). [3]

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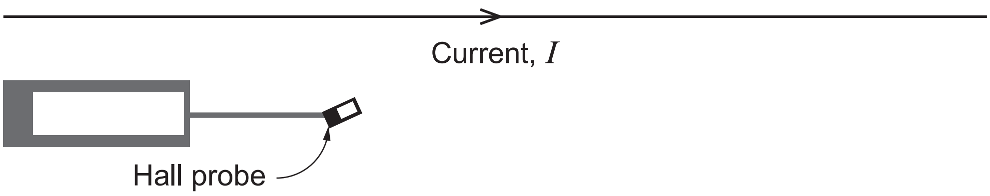
4. *(a)* The current in a long wire is 290 A and is too great to be measured by an ammeter. Explain how you could use a Hall probe, calibrated in tesla (T), to determine this current. [3]



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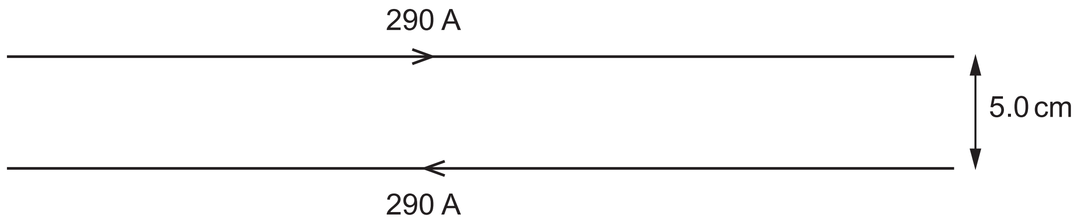
(b) Another long wire carrying the same large current is placed parallel to the original wire as shown. Calculate the force per unit length on each wire also stating the direction of the force on each wire. [4]



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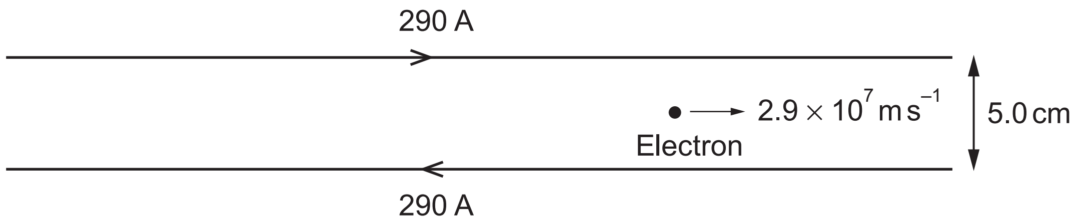
(c) (i) Tirion claims that an electron halfway between the wires, travelling at a speed of $2.9 \times 10^7\,\mathrm{m\,s^{-1}}$ parallel to the wires will perform perfect circular motion between the wires. Determine, using a suitable calculation, whether or not Tirion's claim is correct. The magnetic flux density halfway between the wires is 4.64 mT. [4]



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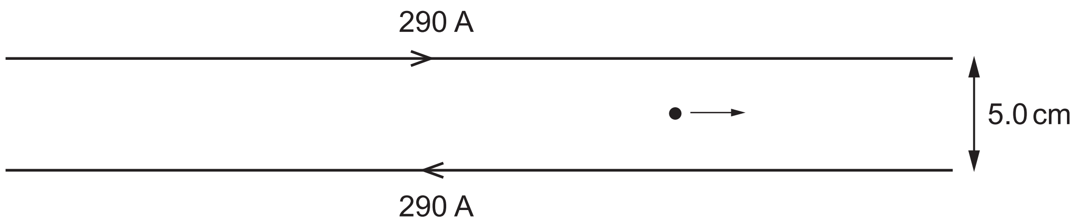
(ii) Sketch the motion of the electron. [2]



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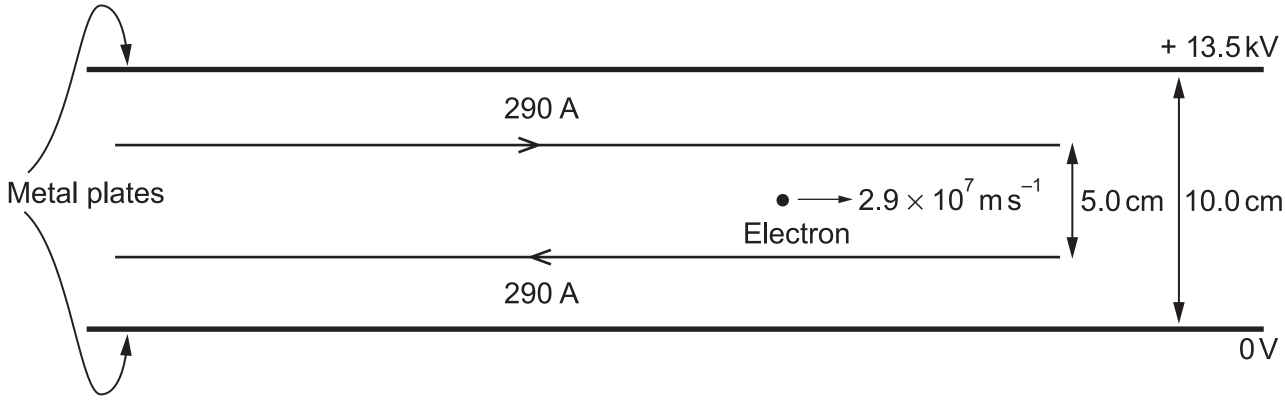
(d) Suppose that an electron travels in a region with a magnetic field and an electric field due to two parallel metal plates as shown below. Deduce whether or not the electron continues with constant velocity (B = 4.64 mT). [5]



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5. *(a)* State the laws of Faraday and Lenz for electromagnetic induction. [2]

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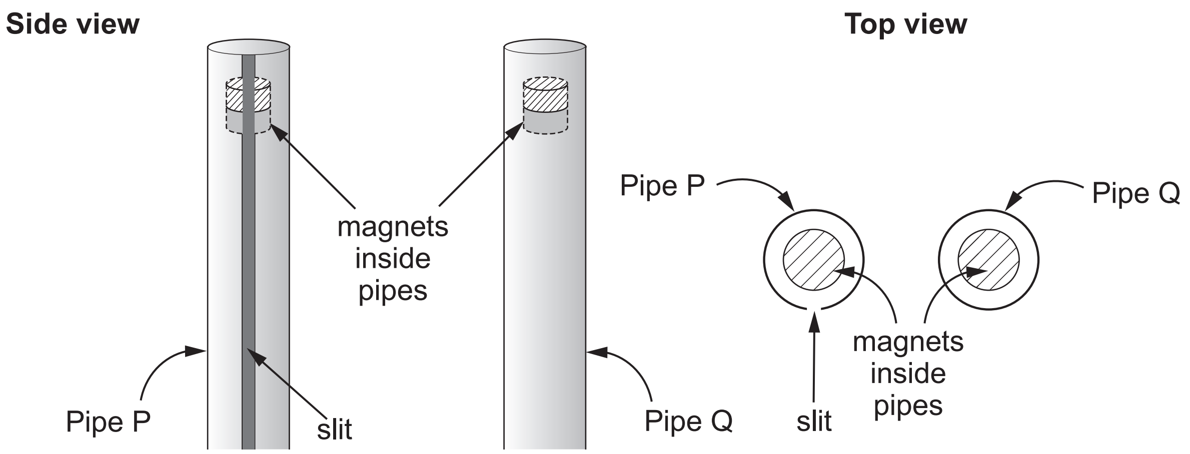
(b) Two strong bar magnets are dropped through two copper pipes (P and Q). Pipe P has a slit running along its length but Q is complete. When the magnet is dropped through pipe P it accelerates almost uniformly but the magnet dropped through pipe Q quickly reaches a very low terminal velocity. Explain these observations. [6 QER]



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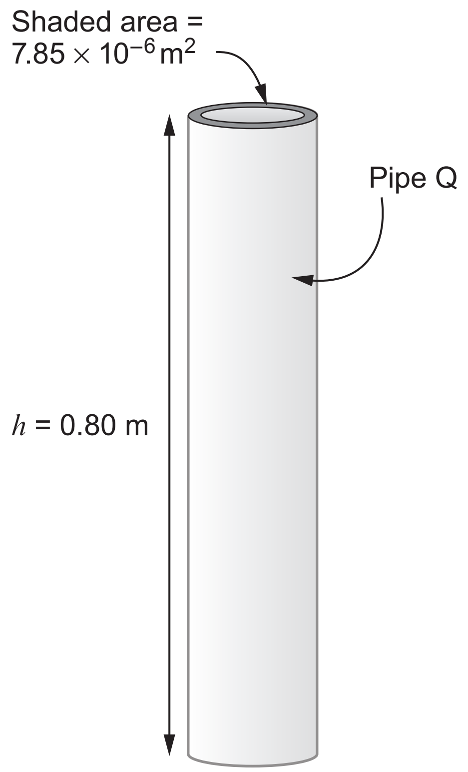
(c) By applying the principle of conservation of energy, calculate the temperature increase of pipe Q after the magnet has fallen at constant speed. [4]

Data:
- mass of magnet = 0.300 kg,
- cross-sectional area of the copper walls of pipe Q = $7.85 \times 10^{-6}\,m^2$ (see diagram)
- density of copper = $8\,960\,kg\,m^{-3}$
- specific heat capacity of copper = $385\,J\,K^{-1}\,kg^{-1}$.



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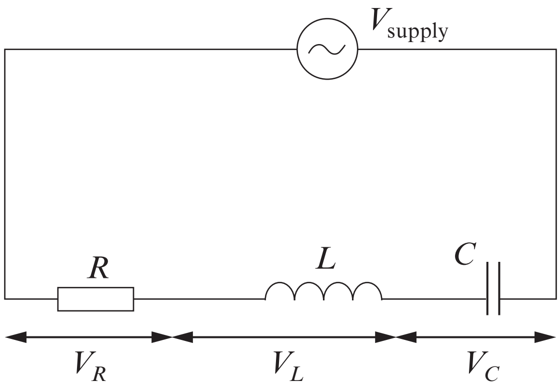
Option A – **Alternating Currents**

6. *(a)* The following RCL circuit is constructed.

variable frequency a.c. supply



For this RCL circuit, describe the relationships between the rms pds V_{supply}, V_R, V_L and V_C:

(i) when the circuit is not in resonance; [3]
Space for diagram.

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(ii) when the circuit is in resonance. [2]

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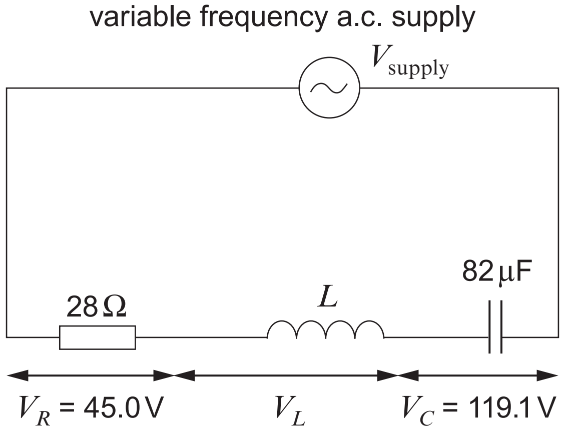
(b) The following RCL circuit is **at resonance**. The values of R and C are provided along with their rms pd values.



Calculate:

(i) the Q factor of the circuit; [1]

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(ii) the rms current; [1]

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(iii) the frequency of the power supply; [2]

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(iv) the inductance of the inductor. [2]

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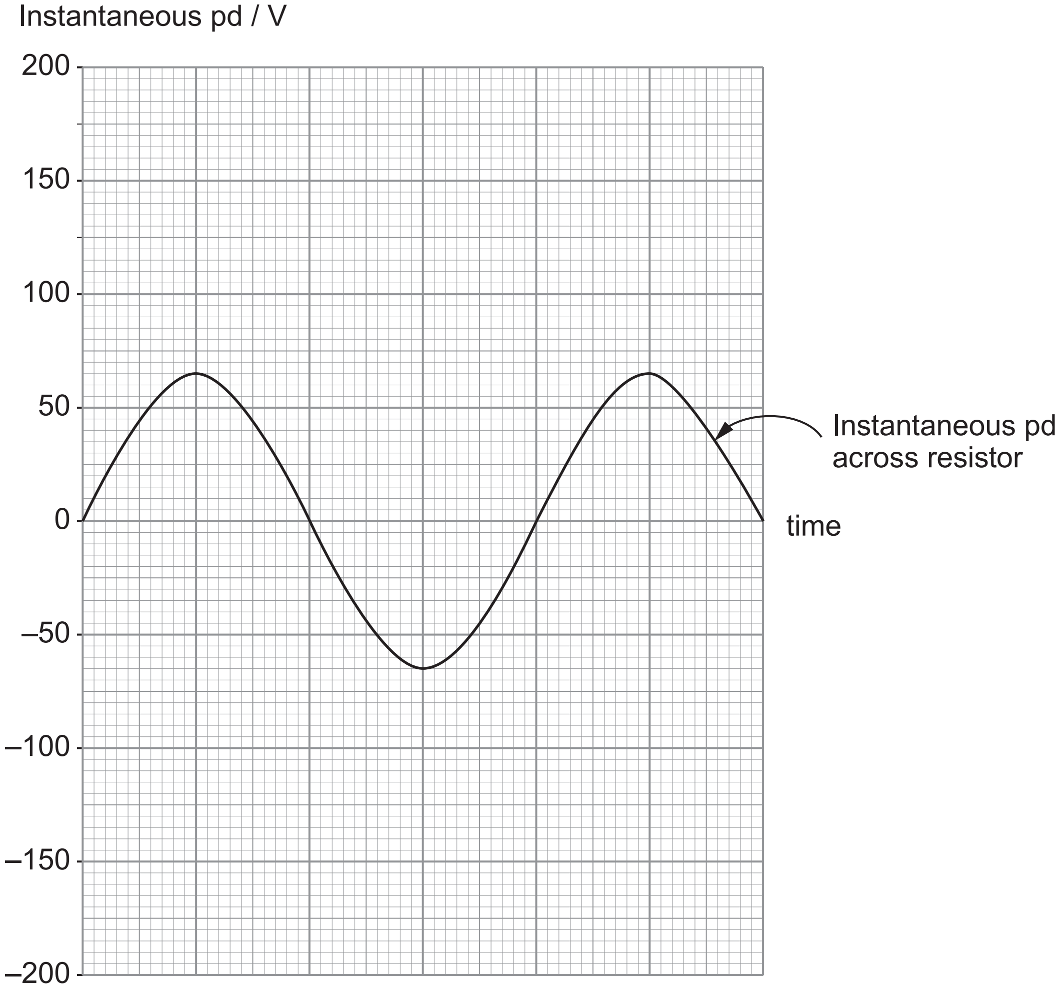
(v) Sketch a graph of the instantaneous pd across the capacitor on the grid provided. (The instantaneous pd across the resistor is shown.) [2]



(vi) Without further calculation, explain why the current decreases when the frequency of the power supply is increased. [2]

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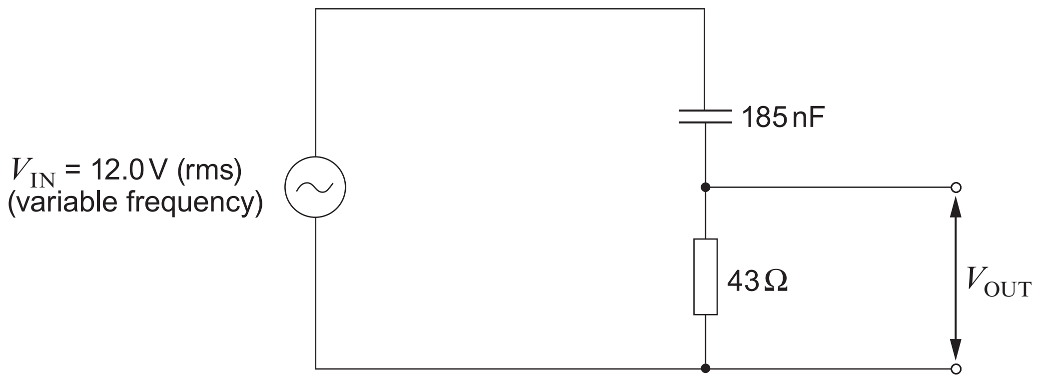
(c) Sam claims that the output pd (V_{OUT}) in the following circuit is greater than 6.0 V when the frequency is greater than 20 kHz, but less than 6.0 V when below 20 kHz. Investigate whether or not Sam is correct. [5]



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Option B – **Medical Physics**

7. *(a)* (i) Discuss how the properties of X-rays make them suitable for medical imaging. [3]

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(ii) An X-ray machine has an operating pd of 18 kV and a current of 12 mA. If only 0.5% of the power is converted into X-rays find:

I. the velocity with which the electrons strike the target; [2]

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II. the power of the emitted X-rays. [2]

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(b) (i) The intensity, I, of an ultrasound beam decreases with the thickness, x, of a material according to the equation:

$$I = I_0 e^{-\mu x}$$

The half-value thickness, $x_{\frac{1}{2}}$ is the thickness of the material that reduces the intensity of an incident beam by 50%. Show that:

$$\mu x_{\frac{1}{2}} = \ln 2$$ [2]

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(ii) The half-value thickness of muscle for ultrasound of frequency 1.0 MHz is 2.7 cm. Determine the thickness of muscle required to reduce the intensity to 70% of its original value. [3]

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(c) A patient has a suspected heart murmur possibly caused by a problem with their aortic valve. You have the choice of the following forms of medical imaging available:

MRI scan **ultrasound B-scan** **fluoroscopy** **CT scan** **X-ray**

Evaluate the effectiveness of each type of imaging in confirming the diagnosis. [5]

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(d) (i) When discussing radiation exposure, medical scientists will mention the *absorbed dose*, D, and the *equivalent dose*, H. Explain the difference between these two terms. [2]

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(ii) During treatment for a cancerous tumour using gamma radiation, a patient's lungs received an equivalent dose of 4 mSv. If the weighting factor of lung tissue is 0.12 calculate the effective dose. [1]

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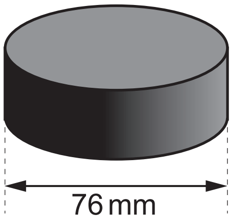
Option C – **The Physics of Sports**

8. This question is about the physics of the motion of an ice hockey puck which is a hard rubber disc of mass 0.17 kg and diameter 76 mm.



(a) When the disc is at room temperature, the coefficient of restitution between the puck and ice is 0.55.

(i) Explain what is meant by the statement *"the coefficient of restitution is 0.55"*. [2]

(ii) When a puck is cooled from room temperature to 0 °C its coefficient of restitution is reduced by 30%. Calculate the bounce height of a puck at 0 °C when it is dropped from an initial height of 0.50 m on to ice. [4]

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(b) The image shows an ice hockey player taking a shot at goal.

(i) When striking the puck, the player changes its speed from $3\,\text{m}\,\text{s}^{-1}$ to $34\,\text{m}\,\text{s}^{-1}$ without changing its direction. The puck remains in contact with the hockey stick for 25 ms. Calculate the mean force exerted by the hockey stick on the puck. [2]

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(ii) The player aims for the top corner of the goal, which is at a height of 1.2 m. The initial angle of the puck's motion is 8° to the horizontal and its speed is $34\,\text{m}\,\text{s}^{-1}$. Determine whether or not the puck ever exceeds the height of the goal.
Ignore the effect of air resistance. [3]

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(iii) The spin rate of the puck at its maximum height is 14 revolutions per second. Calculate its **rotational** kinetic energy.
The moment of inertia of the puck is given by $I = \frac{mr^2}{2}$. [3]

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(iv) Wayne thinks that the answer to part *(b)*(iii) is actually the **total** kinetic energy of the puck at the maximum height. Determine whether Wayne is correct. [2]

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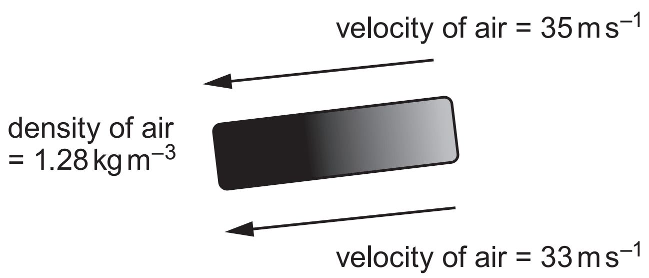
(v) During the shot at goal, the puck is moving to the right. The diagram below shows the velocity of the air relative to the puck. During its flight, the velocity of the air above the puck is greater than that below it creating lift.



Use the Bernoulli equation to calculate the lift force on the puck and show that this is small compared with its weight. [4]

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Option D – **Energy and the Environment**

9. *(a)* (i) State Archimedes' principle. [1]

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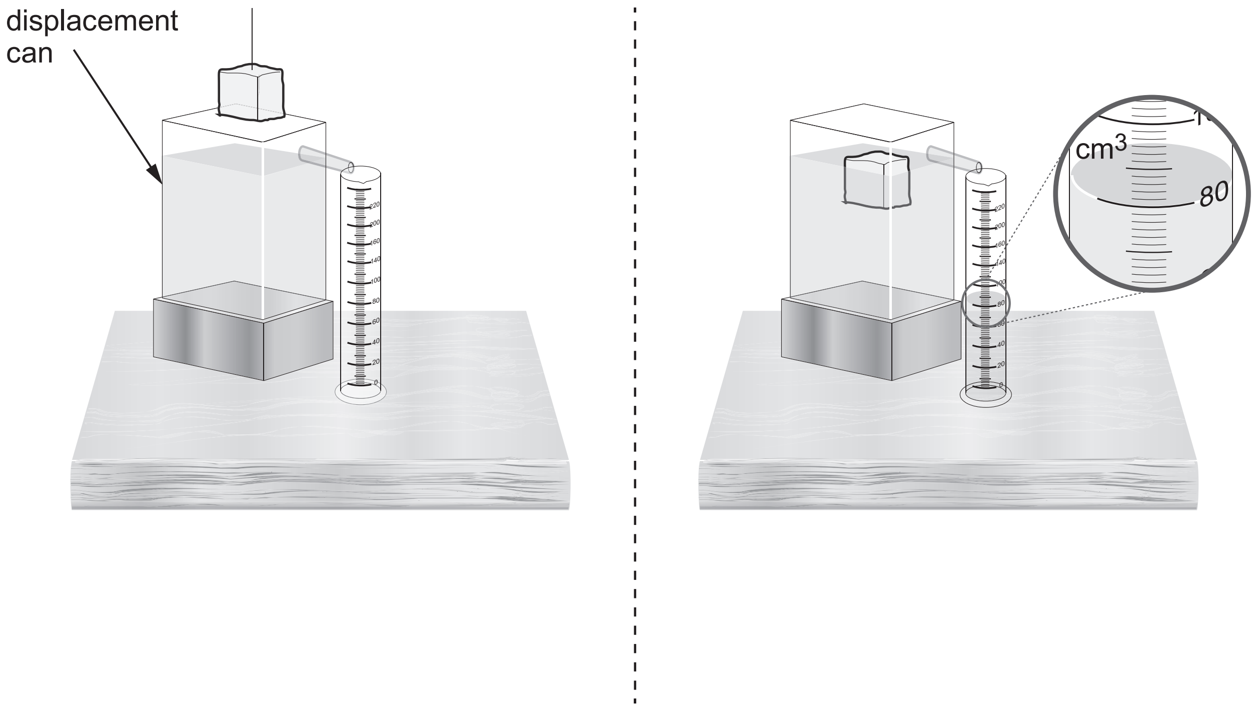
(ii) A student lowers a block of ice into a displacement can containing salt water. As shown in the diagram, 80 cm^3 of displaced salt water is collected in the measuring cylinder.



Calculate the mass of salt water displaced taking the density of salt water, $\rho_{\text{salt water}}$, to be 1 030 kg m^{-3}. [1 cm^3 = 1 × 10^{-6} m^3] [2]

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(iii) The density of ice, ρ_{ice}, is 920 kg m^{-3}. Use this and the Archimedes' principle to determine the volume of the block of ice above the surface. [3]

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(iv) The Greenland ice sheet is a vast body of ice covering roughly 80 % of the surface of Greenland. It is estimated that Greenland's melting ice is releasing 2.2×10^{11} m^3 of water into the surrounding ocean every year.

I. If the surface area of oceans on Earth is 3.6×10^{14} m^2 calculate the rise in water levels per year from Greenland ice melting. [1]

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II. Explain why the melting of the ice sheet would have a greater effect on global sea levels than the melting of the same mass of icebergs. [2]

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III. It is believed that melting ice sheets may lead to further increases in global temperatures. Suggest a reason for this. [1]

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(b) (i) Use an appropriate equation to show that the unit of the coefficient of thermal conductivity, K, is $\text{W m}^{-1}\text{K}^{-1}$. [2]

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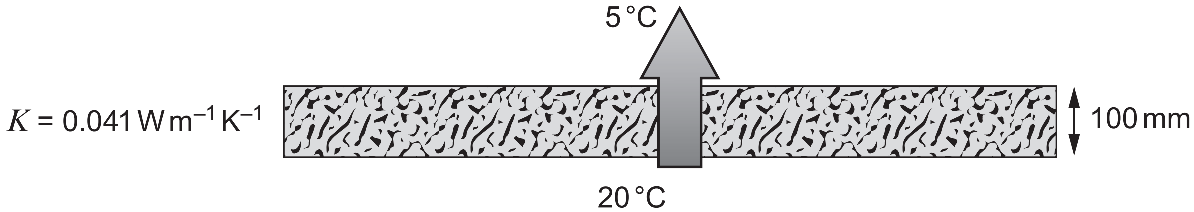
(ii) The recommended thickness of loft insulation has changed over time. In 1985, a 100 mm layer of fibre glass loft insulation ($K = 0.041\,\text{W m}^{-1}\text{K}^{-1}$) was used to cover an area of $72\,\text{m}^2$ in the roof space of a house. During winter, the air temperature just above the insulation was 5 °C and the temperature of the surface supporting the insulation was 20 °C. Calculate the rate of heat flow through the insulation. [2]



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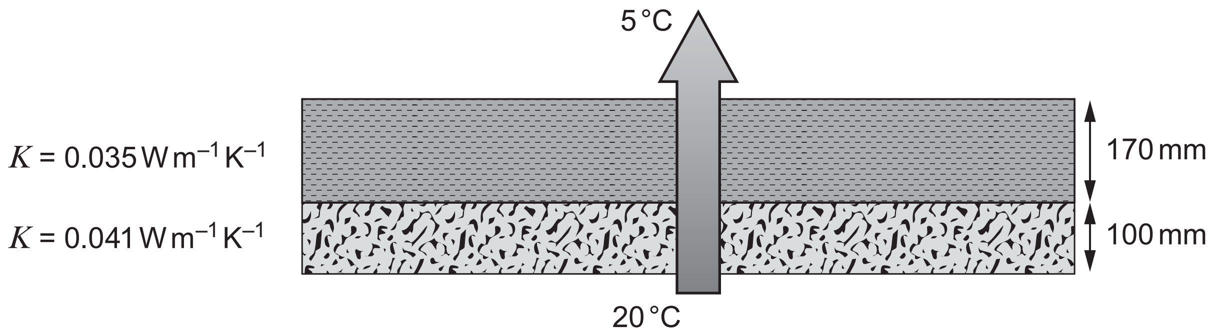
(iii) A present day loft insulation manufacturer recommends a loft insulation thickness of 270 mm. They suggest the house could achieve this modern standard by adding 170 mm of cellulose loft insulation ($K = 0.035\,\text{W m}^{-1}\text{K}^{-1}$) and that it would reduce the rate at which energy is transferred by more than 60 %.



Again, taking the air temperature just above the insulation to be 5 °C and the temperature of the surface supporting the insulation as 20 °C, investigate whether their recommendation is correct. [4]

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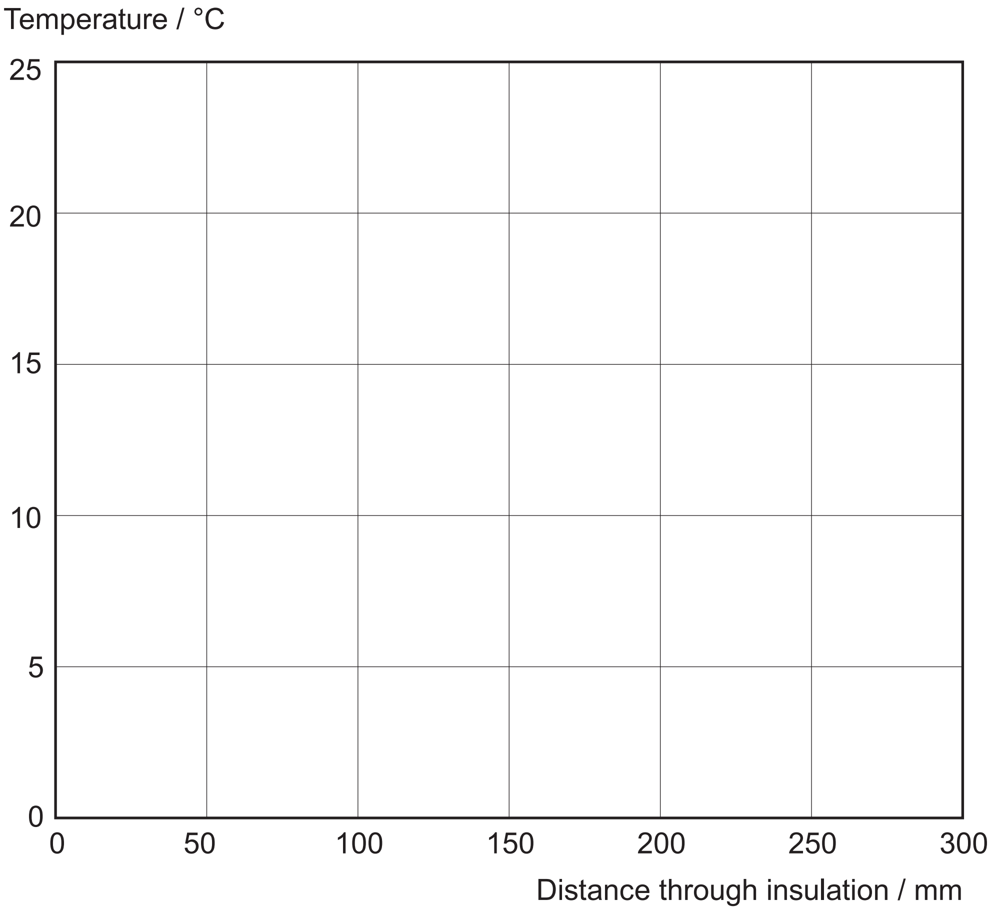
(iv) Starting from the lower surface at 20 °C, sketch a graph of temperature against distance through the loft insulation on the axis provided. [2]



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END OF PAPER

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Surname	Centre Number	Candidate Number
First name(s)		2

GCE A LEVEL

1420U40-1

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FRIDAY, 10 JUNE 2022 – AFTERNOON

PHYSICS – A2 unit 4
Fields and Options

1 hour 35 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**8**	
2.	**9**	
3.	**14**	
4.	**13**	
5.	**22**	
6.	**14**	
Total	**80**	

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ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The total number of marks available for this paper is 80.

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **2**(b).

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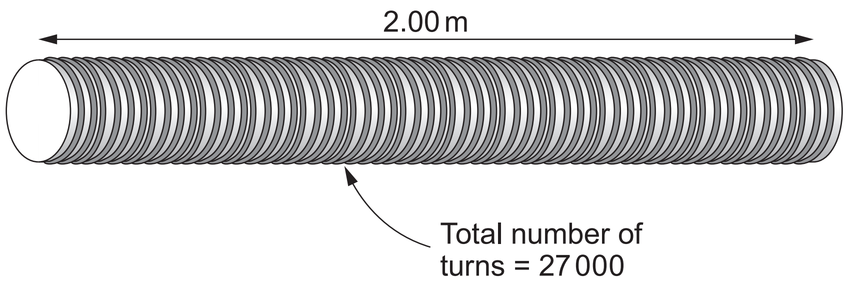
Answer **all** questions.

1. (a) (i) Lindsey calculates that a current of 3.57 A will produce a magnetic flux density of 0.121 T inside the long solenoid shown. Determine whether or not she is correct. [3]



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(ii) State how the magnetic flux density inside the solenoid can be increased greatly without changing the current or the number of turns per unit length. [1]

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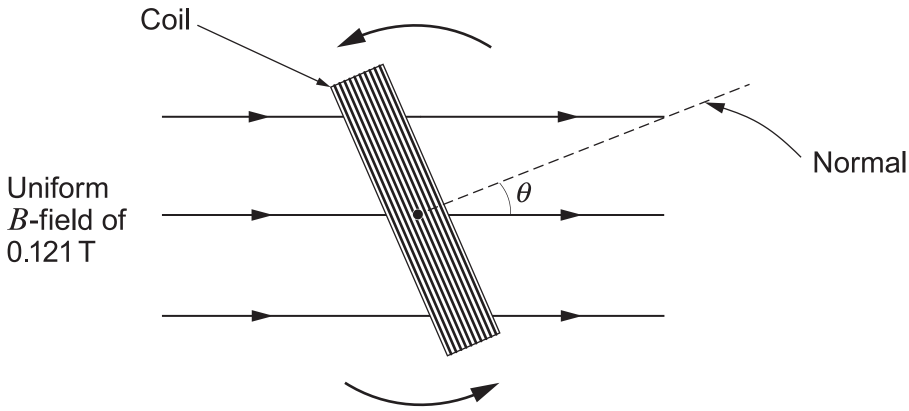
(b) A rectangular coil rotates at a constant angular velocity within a uniform magnetic field of 0.121 T. The coil has 70 turns and cross-sectional area 59 cm^2. The diagram below shows the coil, looking along the axis of rotation.



(i) Calculate the flux linkage of the coil when $\theta = 23°$. [2]

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(ii) As the coil rotates, explain what values of θ provide the maximum **and** minimum values of the induced emf. [2]

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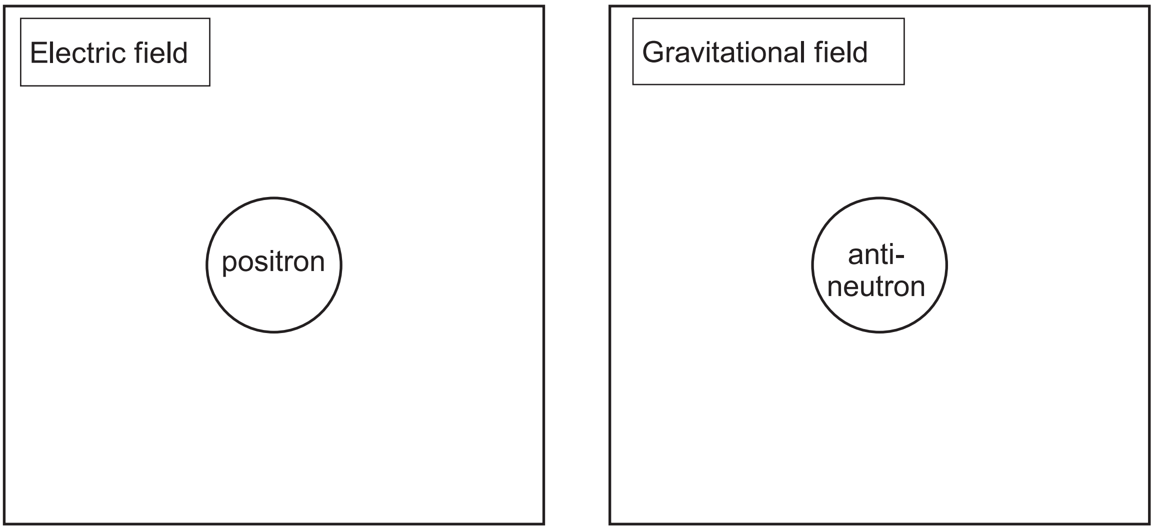
2. (a) Draw 2 equipotentials and 4 field lines for both the **electric** field of the positron and the **gravitational** field of the anti-neutron shown. [3]



(b) Describe and explain the similarities **and** differences between electric **and** gravitational fields. [6 QER]

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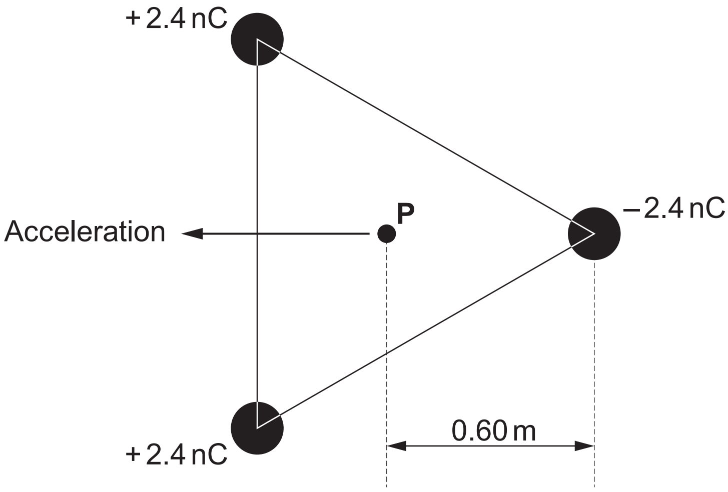
3. Three stationary charges are located at the corners of an **equilateral** triangle and an electron is located at the centre, **P**, of the triangle as shown. The electron is exactly 0.60 m from each charge.



(a) Draw arrows to show the 3 electric fields at **P** due to the 3 charges. Use these to explain why the electron will accelerate in the direction shown. [4]

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(b) Calculate the initial acceleration of the electron. [4]

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(c) Show **clearly** that the initial potential energy of the electron is approximately -6×10^{-18} J. [2]

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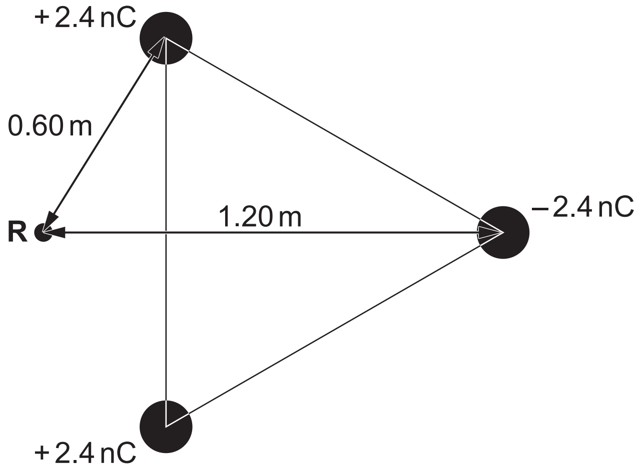
(d) Determine whether or not the electron (initially at rest) will reach point **R**, shown in the diagram below (both positive charges are 0.60 m from **R**). Explain your reasoning. [4]



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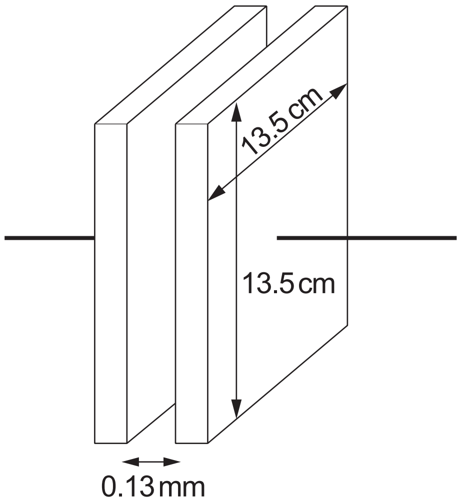
4. (a) Calculate the capacitance of the capacitor shown. [2]



(b) (i) Calculate the pd across the terminals of a 5.0 mF capacitor when it stores 1.00 J of energy. [2]

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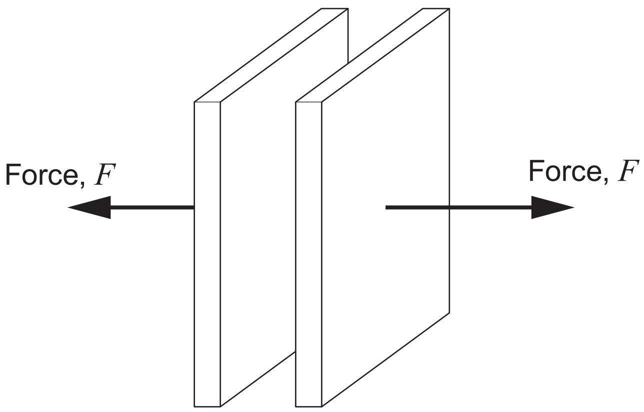
(ii) In the space provided, draw a diagram to show how you would combine three 5.0 mF capacitors to produce a capacitance of 7.5 mF. [2]

(c) The separation of the plates of a charged capacitor is increased by the application of a force, F. The capacitor is **isolated** so the charges on the plates remain unchanged.



(i) State why a force must be exerted to separate the charged plates. [1]

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(ii) Explain why the capacitor stores more **energy** when the separation of the plates is increased even though the charge remains constant. [2]

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(iii) Show that the energy stored by the capacitor is given by:

$$\text{energy stored} = \frac{1}{2}\frac{Q^2 d}{\varepsilon_0 A}$$

where Q is the charge stored, d is the separation of the plates, ε_0 is the permittivity of free space and A is the area of the plates. [2]

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(iv) Bethan states that the force, F, required to separate the plates is given by:

$$F = \frac{1}{2}\frac{Q^2}{\varepsilon_0 A}$$

Determine whether Bethan is correct to arrive at this conclusion. [2]

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5. (a) (i) Derive the expression for the critical density of a flat universe. [4]

$$\rho_c = \frac{3{H_0}^2}{8\pi G}$$

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(ii) Use this equation to show that the critical density of the universe corresponds to approximately 5 hydrogen atoms per m^3. [2]

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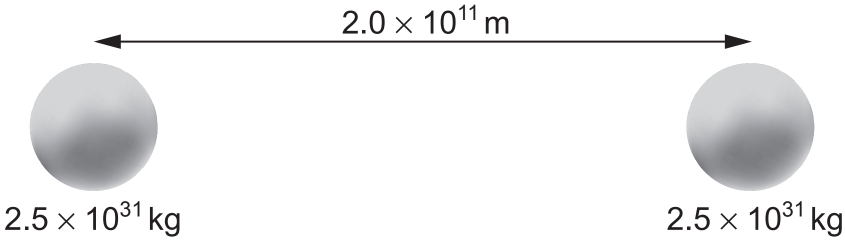
(b) A two-star system has two stars of equal mass as shown.



(i) State or calculate the position of the centre of mass of the two-star system. [1]

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(ii) Calculate the period of orbit of the stars about their centre of mass. [2]

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(iii) Calculate the maximum red shift (or blue shift) of the hydrogen 434 nm line due to the orbital motion of the stars. [3]

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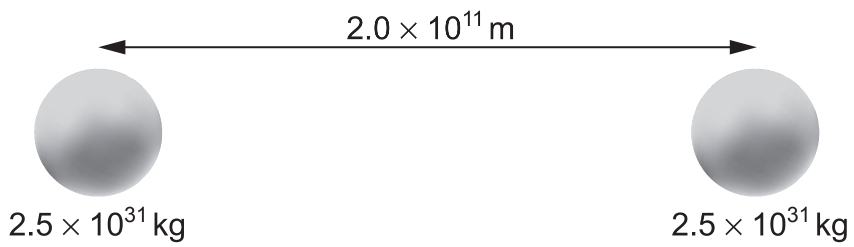
(c) The centre of mass of this two-star system is also the point where the resultant gravitational field strength is zero. Explain why these points only coincide when the stars have identical masses. [2]



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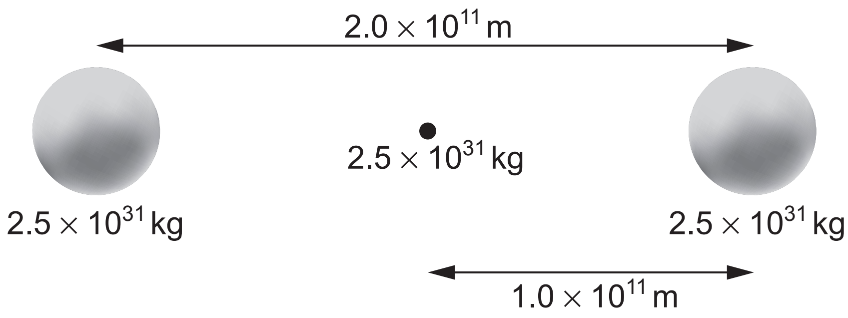
(d) A different star system consists of three stars all of equal mass. Two stars orbit around a stationary black hole and the black hole is always halfway between the two stars as shown.



(i) Explain why the resultant force on the black hole is always zero. [1]

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(ii) Explain why the gravitational force acting on either of the orbiting stars is five times greater in this three-star system than the two-star system of part (b). [2]

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(iii) Joseff claims that the stars in the three-star system will provide a red shift that is five times larger than that of the two-star system of part (b). Evaluate whether or not he is correct. [3]

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(e) A recent theoretical publication suggests that the decay of the Higgs Boson will give direct evidence for dark matter. Suggest what needs to be done for this theory to be generally accepted by scientists in the future. [2]

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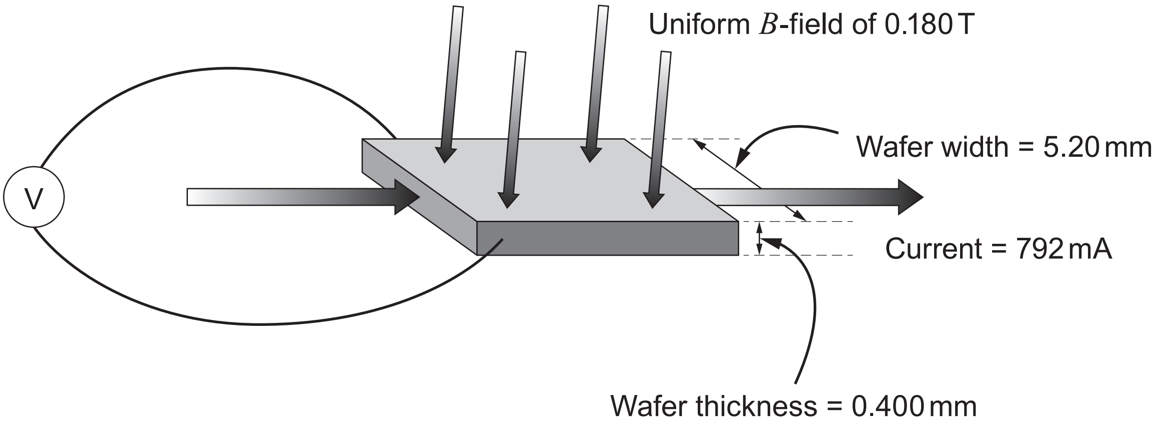
6. (a) Catrin carries out a Hall effect experiment to find out the number of free electrons per unit volume in a certain metal. She places a wafer of the metal in a known magnetic field and passes a current through it (see diagram).



(i) By considering the forces acting on free electrons passing through the wafer, explain why a Hall voltage is measured on the voltmeter shown. [4]

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(ii) By equating the magnetic and electrical forces, show that the Hall voltage, V_H, is given by:

$$V_H = Bvd$$

where B is the magnetic flux density, d is the width of the wafer and v is the drift velocity. [3]

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(iii) Catrin states that her measured Hall voltage of 68.0 nV is consistent with a drift velocity of approximately $0.07\,\text{mm}\,\text{s}^{-1}$. Determine whether, or not, she is correct. [2]

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(iv) Calculate the number of free electrons per unit volume for the metal. [3]

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(b) After repeating the same experiment on a different metal, Catrin obtains a value of $(5.85 \pm 0.19) \times 10^{28}\,\text{m}^{-3}$, for the number of free electrons per unit volume. She is given a table of values in order to determine which metal has been used in the experiment.

Element	Free electron density / $10^{22}\,\text{cm}^{-3}$
Aluminium	18.1
Barium	3.15
Copper	8.47
Gold	5.90
Iron	17.0
Silver	5.86

Explain which metal(s) she should conclude has been used in the experiment. [2]

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END OF PAPER

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Surname	Centre Number	Candidate Number
First name(s)		2

GCE A LEVEL

S23-1420N40-1-R1

1420U40-1

THURSDAY, 15 JUNE 2023 – MORNING

PHYSICS – A2 unit 4
Fields and Options

2 hours

	For Examiner's use only		
	Question	Maximum Mark	Mark Awarded
Section A	1.	11	
	2.	14	
	3.	19	
	4.	17	
	5.	10	
	6.	9	
Section B	Option	20	
	Total	100	

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Answer **all** questions.

Write your name, centre number and candidate number in the spaces at the top of this page.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

This paper is in 2 sections, **A** and **B**.

Section **A**: 80 marks. Answer **all** questions. You are advised to spend about 1 hour 35 minutes on this section.

Section **B**: 20 marks. Options. Answer **one option only**. You are advised to spend about 25 minutes on this section.

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **5(b)**.

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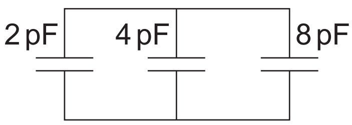
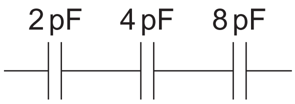
SECTION A

Answer **all** questions.

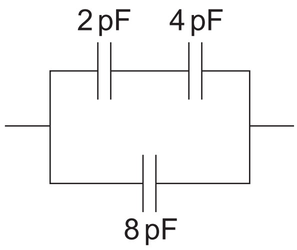
1. (a) Calculate the total capacitance of each of the following capacitor combinations. [6]



(i)



(ii)



(iii)

(i) ....

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(ii) ....

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(b) Michael, a laboratory technician, must make a 2.7 μF capacitor using only two square sheets of aluminium of side 42.0 cm. Determine whether or not it is realistic for Michael to produce a capacitance of 2.7 μF using these two sheets of aluminium and no dielectric. [4]

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(c) State what happens to the capacitance of a parallel plate capacitor when a dielectric is placed between the plates. [1]

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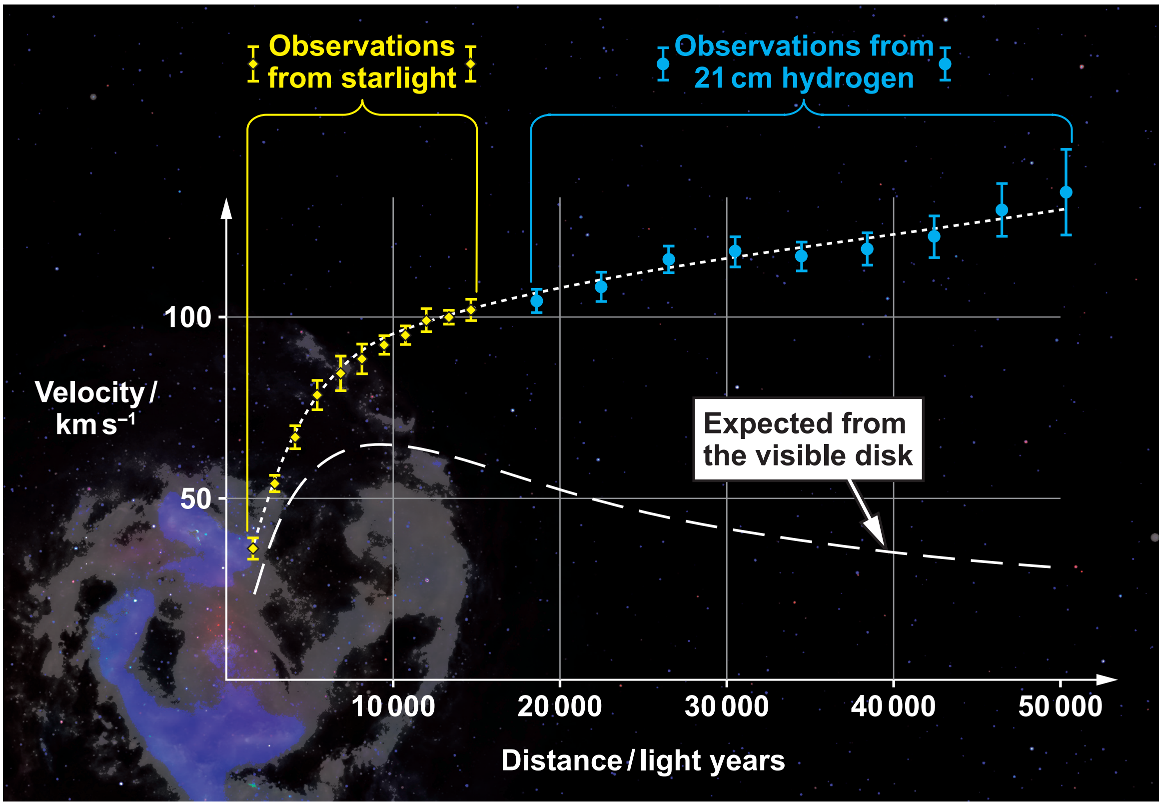
2. Light from a spiral galaxy is analysed and the following graph of velocity against distance from the centre of the galaxy is obtained.



(a) Show that the speed, v, of an object in a circular orbit of radius, r, about a massive object of mass, M, is given by: [3]

$$v = \sqrt{\frac{GM}{r}}$$

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(b) Hence, explain why the graph is considered to be evidence for dark matter. [3]

(c) The lower curve in the graph suggests that an object a distance of 20 000 light years from the centre of the galaxy should have an orbital speed of approximately 50 km s^{-1}. Use this data to estimate the visible mass of the galaxy. (1 light year = 9.46×10^{15} m) [3]

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(d) The observed data at 30 000 light years from the galactic centre were obtained using microwaves of wavelength 21 cm. Calculate the approximate wavelength shift that was observed to obtain the data plotted at 30 000 light years. [3]

(e) Use the Hubble equation to calculate the distance, from Earth, for a galaxy to have the same recessional speed as that of part (d). [2]

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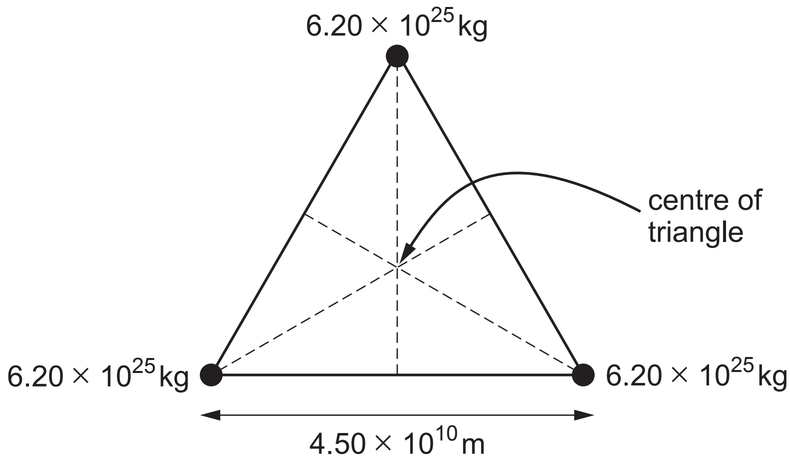
3. Three planets of equal mass are arranged in an **equilateral triangle** as shown.



(a) (i) Show that the gravitational force exerted by one planet on another is approximately 1.3×10^{20} N. [2]

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(ii) Hence, explain why the resultant gravitational force experienced by any of the planets is approximately 2×10^{20} N towards the centre of the triangle. [4]

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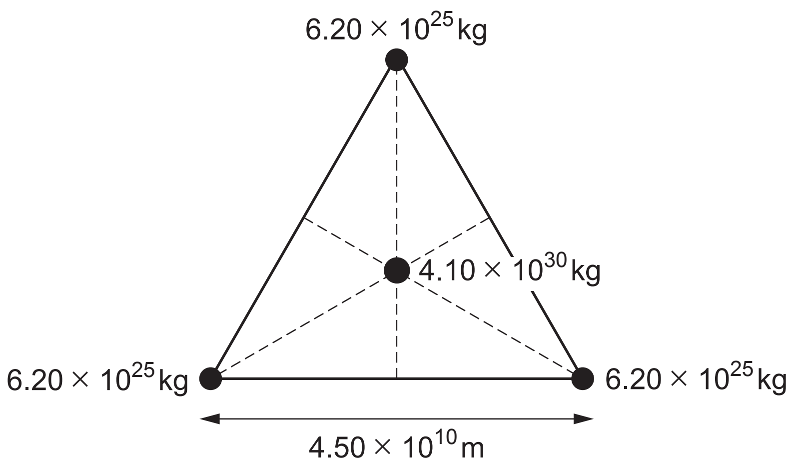
(b) A star is located at the centre of the triangle.



(i) By adding to the diagram, show that the distance between the star and any of the planets is given by: [2]

$$\frac{4.5 \times 10^{10}}{2 \times \cos 30°}$$

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(ii) Rhodri states that the force calculated in part (a)(ii) is negligible compared with the gravitational force exerted on a planet by the star. Determine whether or not Rhodri is correct. [2]

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(c) (i) Show that the gravitational potential at the position of a planet due to the star and the other two planets is approximately $-1 \times 10^{10}\,\mathrm{J\,kg^{-1}}$. [3]

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(ii) Rhodri now states that, if an object with kinetic energy 7×10^{35} J struck one of the planets, the planet would gain enough energy to escape the gravitational pull of the star. Discuss to what extent Rhodri's statement might be correct. [3]

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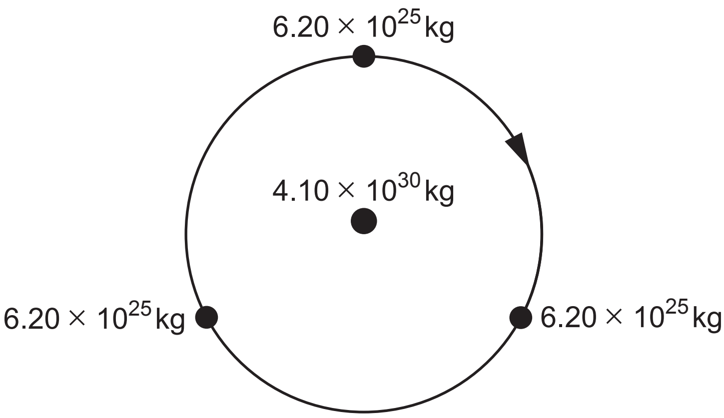
(d) The three planets, whilst orbiting the star, stay in an equilateral triangle. Delyth claims that these planets cannot be detected by the periodic Doppler shift of the star. Determine whether or not Delyth is correct. [3]



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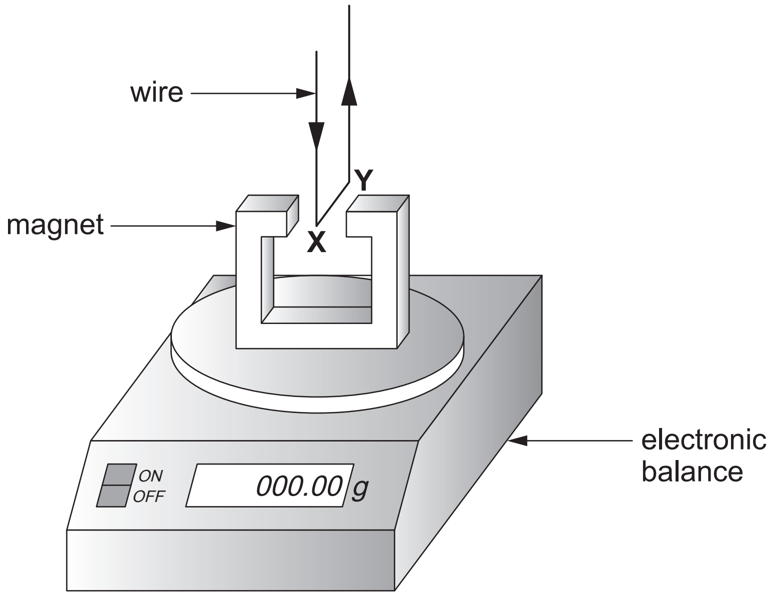
4. The following set-up is used to investigate the force on a current-carrying wire in a magnetic field.



The part **XY** of the wire which is in the magnetic field, has been placed carefully so that the current is from front to back at 90° to the direction of the magnetic field. When there is no current the balance is reset to display 000.00 g. When there is a current in the direction shown, the magnetic force results in a positive reading on the display of the balance. However, this signifies an upward force on the wire.

(a) (i) Explain briefly why the magnetic force must be upward on the part **XY** of the wire. [2]

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(ii) **On the diagram**, indicate which is the North pole of the magnet, **and** state below which rule you used to obtain your answer. [2]

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(iii) Explain why the vertical parts of the wire have no effect on the reading displayed by the electronic balance. [1]

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(b) Bronwen varies the length of XY and records the electronic balance reading each time when a current of 5.00 A flows. She records all her results in a table and plots a graph of magnetic force against length of wire.

(i) **Complete the table and plot** the two missing points on the graph. [4]

Length of XY / cm	Balance reading / g	Magnetic force / 10^{-3} N
1.0	0.56	5.5
2.0	1.22	....
3.0	1.75	17.2
4.0	2.36	23.1
5.0	2.89	28.4
6.0	3.32	32.6
7.0	....	35.8

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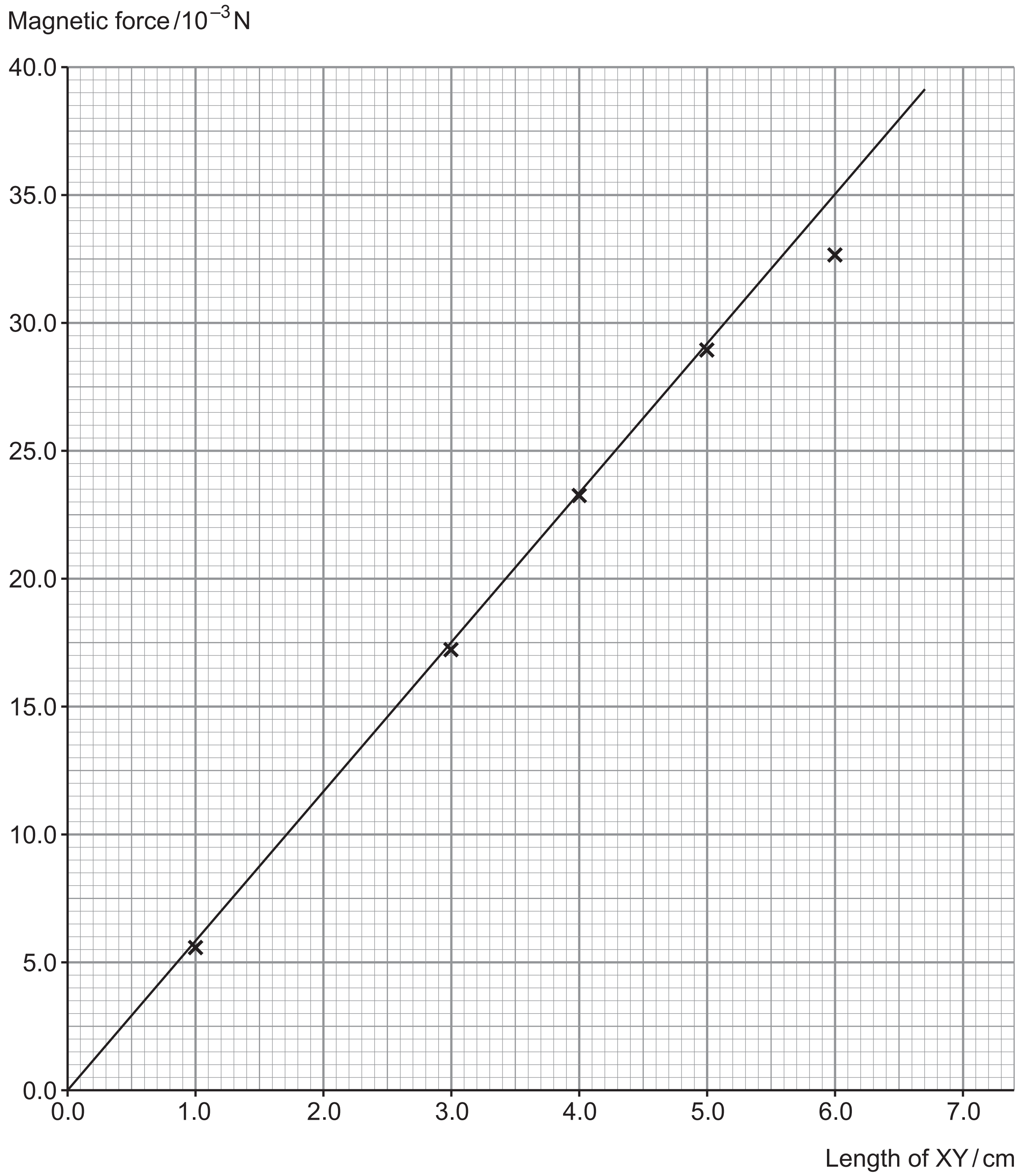

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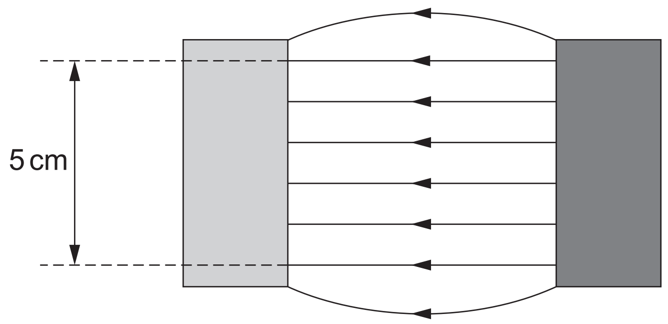
(ii) The region of uniform magnetic field between the poles of the magnet is of length 5 cm only. Evaluate whether or not Bronwen has drawn the line of best fit correctly. [3]



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(iii) Determine the magnetic flux density, B, of the uniform magnetic field between the poles of the magnet and quote it to an appropriate number of significant figures. The current in the wire is 5.00 A. No uncertainty is required in your value of B. [5]

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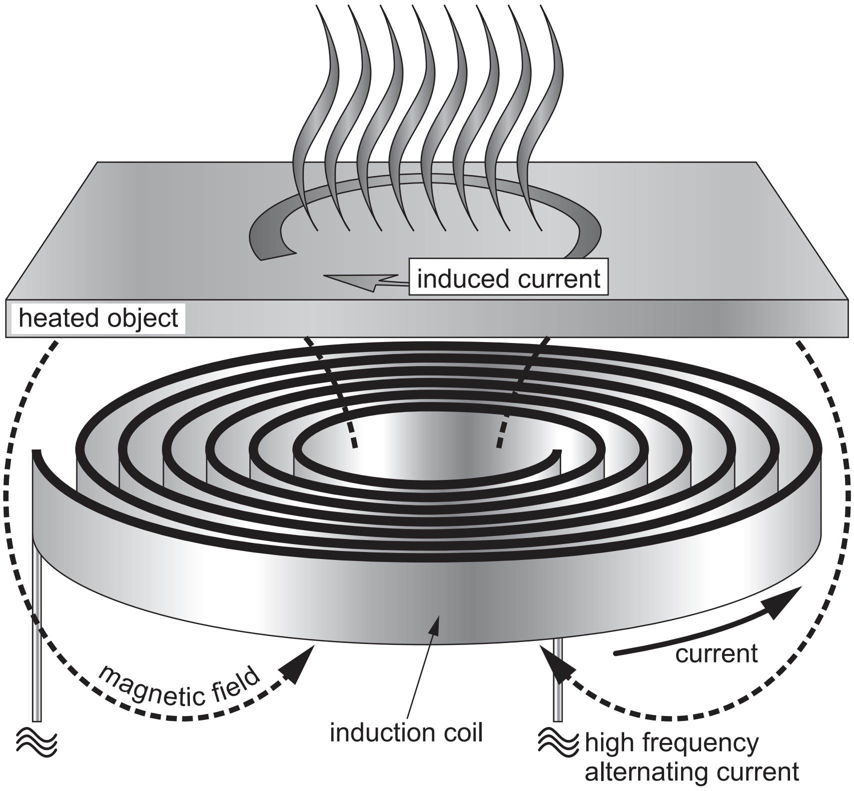
5. (a) The diagram shows how conductors can be heated using a varying magnetic field.



Explain why the heated object gets hot **and** why a high frequency alternating current is used. [4]

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(b) Explain how a Hall voltage arises in a Hall probe, how it is measured **and** how this can be used to measure the magnetic flux density. A diagram should be included in your answer. [6 QER]

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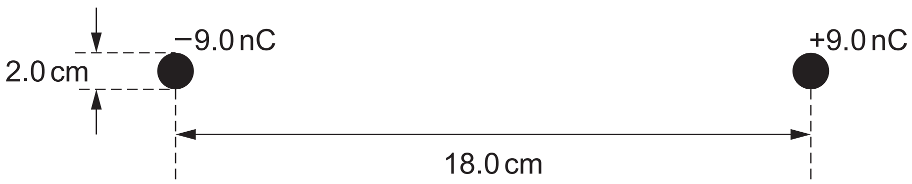
6. Two small spheres of radius 1.0 cm carry uniform charges of +9.0 nC and –9.0 nC. Their centres are separated by a distance of 18.0 cm as shown. Treat the charges as being at the centres of the spheres.



(a) (i) Show that the potential at the surface of the negative sphere is approximately –7500 V. [3]

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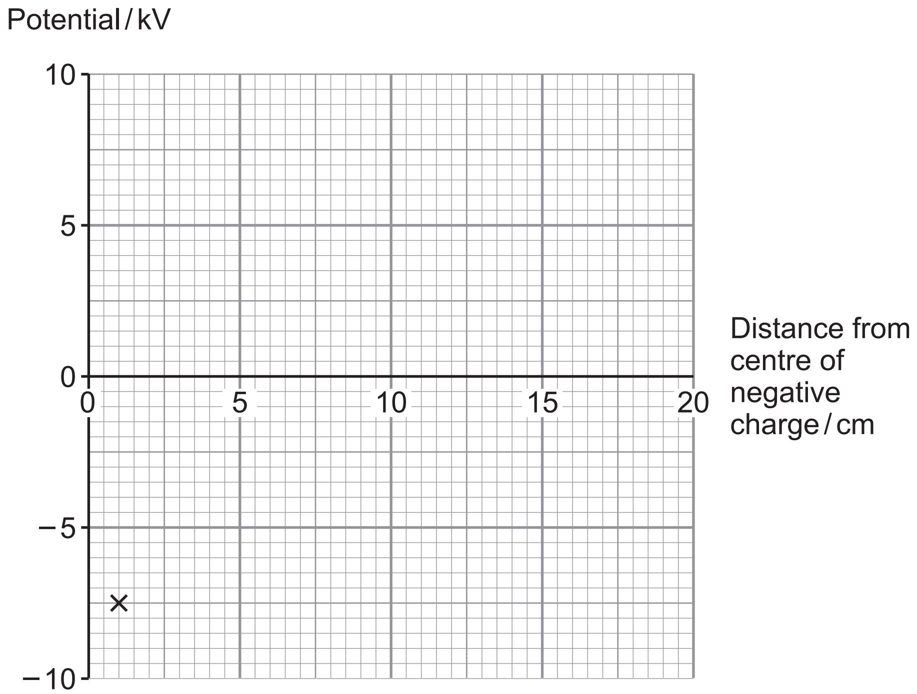
(ii) Sketch the graph for the electric potential between the two charges (the potential has been plotted for the surface of the negative charge). [3]



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(b) Explain why the electric field on the right-hand surface of the negative sphere is approximately 810 000 N C^{-1} and directed towards the left. [3]

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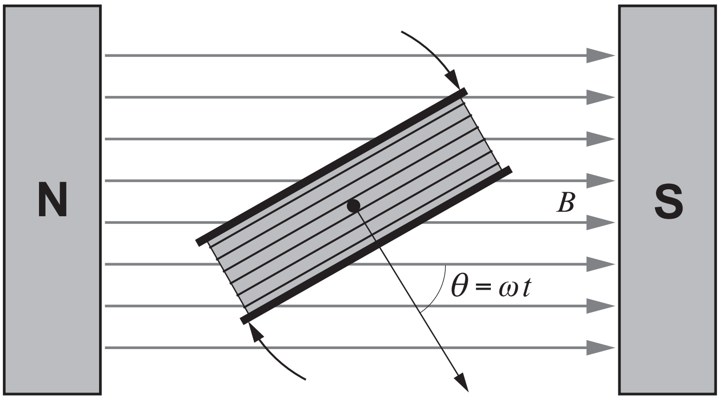
Option A – **Alternating Currents**

7. (a) A diagram of a rotating coil in a magnetic field is shown below.



The coil rotates 50 times per second in a field of flux density 0.215 T.
The coil has 2 650 turns and cross-sectional area $2.35 \times 10^{-3}\,m^2$. (Note that $\theta = 0$ when $t = 0$)

(i) Calculate the peak induced emf in the coil. [2]

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(ii) Calculate the flux linkage of the coil when $t = 5.0\,ms$.
[Hint: put your calculator in radian mode.] [2]

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(iii) Calculate the induced emf in the coil when $t = 5.0\,ms$. [1]

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(iv) Vanessa states that her answers to parts (ii) and (iii) are consistent with the coil having completed a quarter of a cycle. Determine whether or not she is correct. [3]

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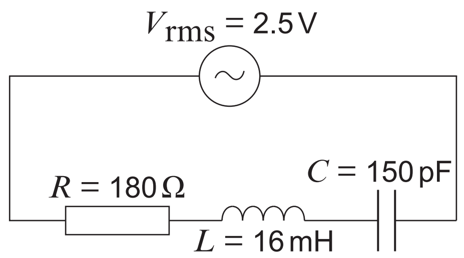
(b) (i) Calculate the resonance frequency of the following circuit. [2]



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(ii) Explain why the rms current at resonance is approximately 14 mA. [2]

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(iii) Calculate the rms current in the circuit when the frequency is 105 kHz. [4]

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(iv) Explain why your answers to parts (b)(i), (b)(ii) and (b)(iii) suggest that the circuit has a high Q factor. [2]

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(v) Thomas considers the effect, on this circuit, of increasing the capacitance only. He believes that this will decrease the Q factor. Determine whether or not he is correct. [2]

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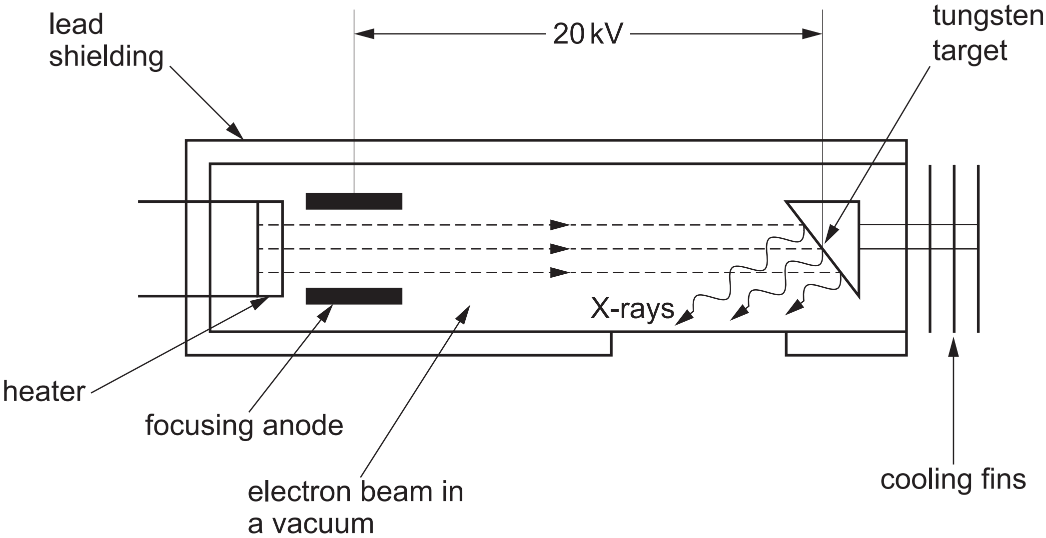
Option B – **Medical Physics**

8. (a) Below is a simplified diagram of the inside of an X-ray tube.



(i) State the main function of the heater **and** explain why there must be a vacuum between the heater and the tungsten target. [2]

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(ii) The electron beam has a current of 100 mA. Determine how many electrons arrive at the tungsten target per second. [2]

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(iii) Estimate the velocity of an electron in the X-ray tube as it hits the target. [3]

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(iv) Determine the minimum wavelength, λ_{min}, of the X-rays produced by this tube. [3]

(b) (i) An ultrasound probe can be used to study the speed of blood flow from the heart. Explain how the probe produces ultrasound **and** how the speed of blood flow can be determined. [3]

(ii) In measuring the speed of blood flow, the frequency of ultrasound used is 2.0 MHz and it travels through the blood at $1570\,\mathrm{m\,s^{-1}}$. A frequency shift of 0.23 kHz is measured when the ultrasound is incident at an angle of 37° to the blood flow. Calculate the speed of the blood flow. [2]

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(c) Doctors are concerned that a patient is suffering from an overactive thyroid gland (hyperthyroidism). They have the choice of the following techniques to help diagnose this.

X-ray **MRI** **ultrasound** **CT scan** **radioactive tracers**

Evaluate the suitability of **all five** types of imaging techniques for detecting an overactive thyroid gland. [5]

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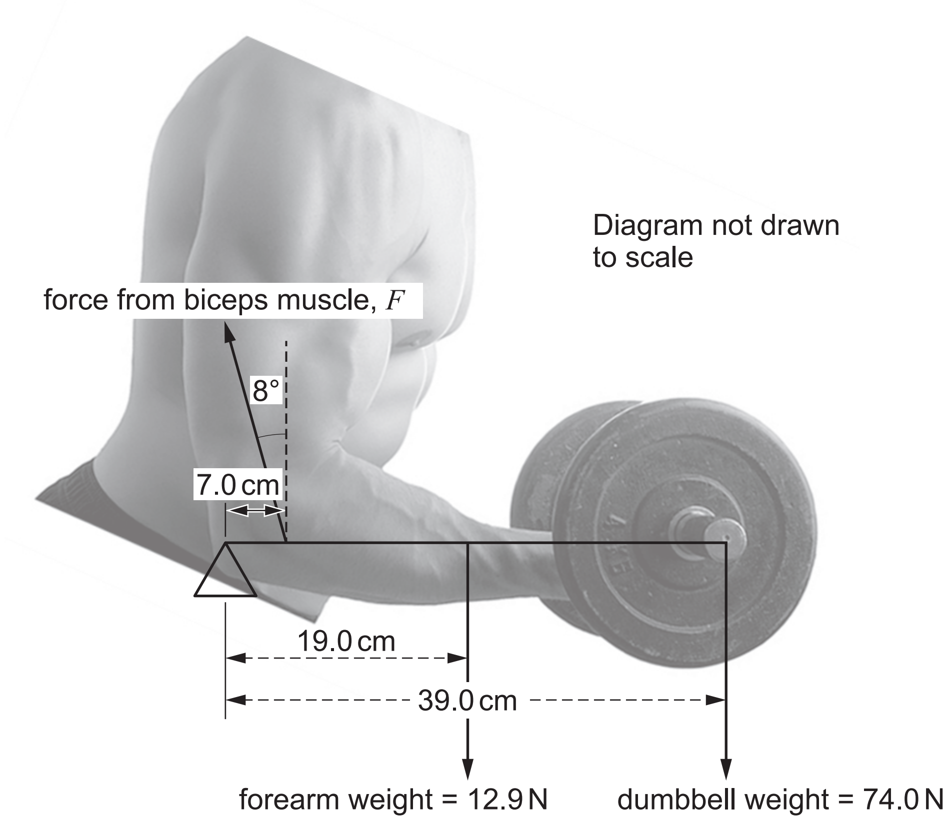
Option C – **The Physics of Sports**

9. (a) An athlete lifts a dumbbell of weight, 74.0 N as shown in the following diagram. Calculate the magnitude of the force, F, that the biceps muscle exerts on the athlete's forearm assuming that the elbow can be modelled as a pivot. [2]



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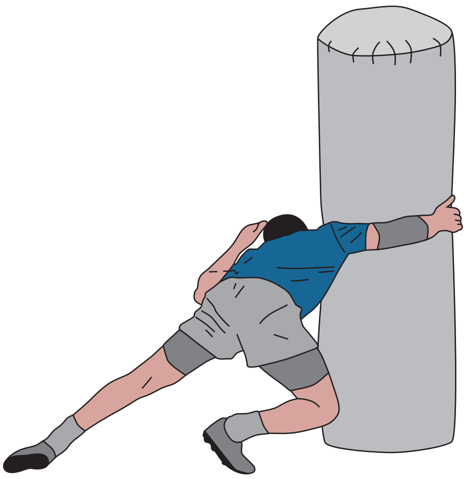
(b) A rugby player of mass 117 kg runs at a tackle pad with an initial velocity 7.75 $m s^{-1}$. In the collision with the tackle pad, the player experiences a mean force of 5500 N for a contact time of 0.176 s. This force is in the exact opposite direction to the initial velocity. Calculate the final velocity of the rugby player after the collision. [3]



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(c) (i) During a rugby match, a player wishes to kick the ball as far as possible. Describe the factors that affect the drag force on the ball. You may use the following diagrams to support your answer. [3]

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(ii) A kicking coach thinks that, if the number of revolutions per second of the ball is doubled, then the rotational kinetic energy of a rugby ball will also double. Evaluate whether or not the coach is correct. [2]

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(iii) The rugby ball is dropped on to two playing surfaces, A and B. Evaluate whether the playing surfaces are different. [3]

Playing surface	Initial height / m	Bounce height / m
A	7.8	1.2
B	4.5	0.7

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(iv) When a rugby ball is kicked it gains an angular velocity of 37.7 rad s^{-1} in a time of 17.0 ms. The moment of inertia of the rugby ball is 0.0028 kg m^2. Calculate the torque exerted on the ball by the kicker's foot. [2]

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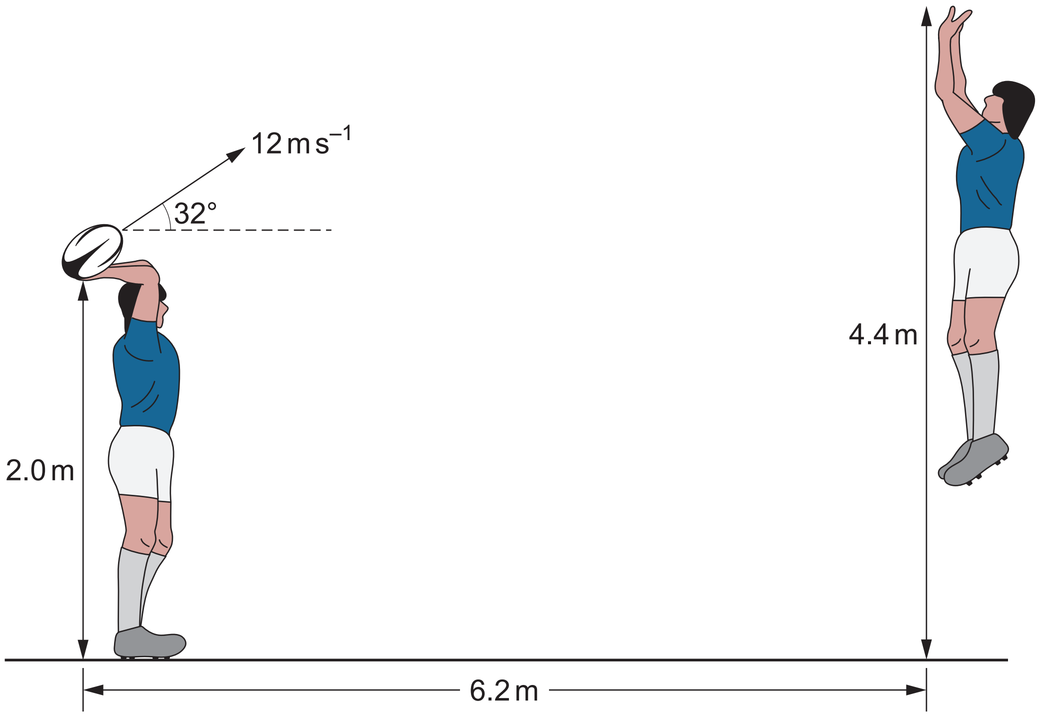
(v) A player throws a ball from a height of 2.0 m with an initial speed of 12 m s^{-1} at an angle of 32° to the horizontal. The maximum height reached by the player catching the ball is 4.4 m and he is standing 6.2 m away from the thrower. Explain why it is possible for the catcher to catch the ball. [5]



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Option D – **Energy and the Environment**

10. (a) (i) The mean temperature of the Earth is 288 K. Assuming it behaves as a black body, show that the wavelength of the peak emission of radiation is in the infra-red region of the electromagnetic spectrum. [2]

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(ii) Carbon dioxide, methane and water vapour are greenhouse gases. Describe the role these gases play in the greenhouse effect. [3]

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(b) (i) The Greenland ice sheet is the second largest ice sheet in the world. Over the last two decades the mean decrease in volume of Greenland's ice is $3.0 \times 10^{11}\,\text{m}^3\,\text{year}^{-1}$. Estimate the volume of water released into the surrounding ocean in this time period. [$\rho_{\text{ice}} = 920\,\text{kg}\,\text{m}^{-3}$, $\rho_{\text{water}} = 1\,000\,\text{kg}\,\text{m}^{-3}$] [2]

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(ii) Satellite images show that the edge of the Greenland ice sheet appears brown in places. Scientists report that this discolouration is due to microscopic plants flourishing in the melting ice. Explain the effect this could have on the rate at which the ice melts. [2]

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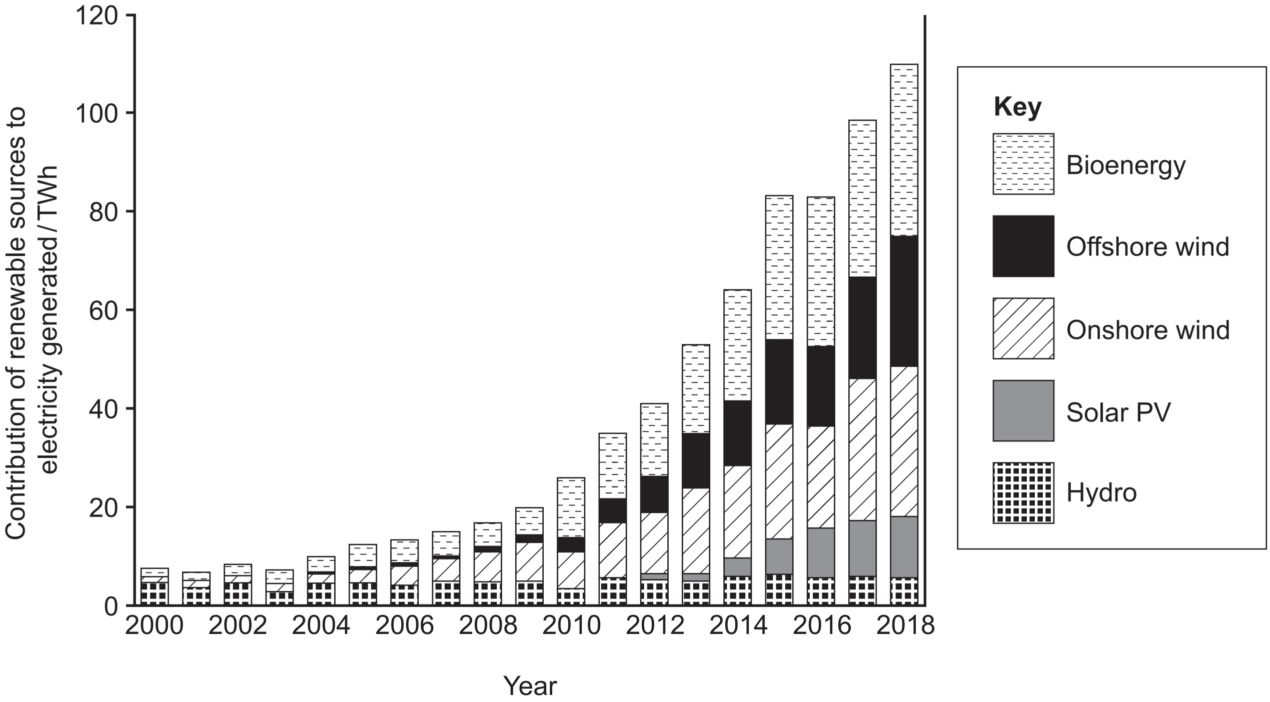
(c) The graph shows how renewable sources have contributed to the electricity generated in the UK between 2000 and 2018.



Note: Hydro bar includes shoreline wave/tidal (0.009 TWh in 2018)

(i) Compare the contribution of solar photovoltaic (Solar PV) with hydroelectric power since 2010 **and** suggest a reason for this difference. [2]

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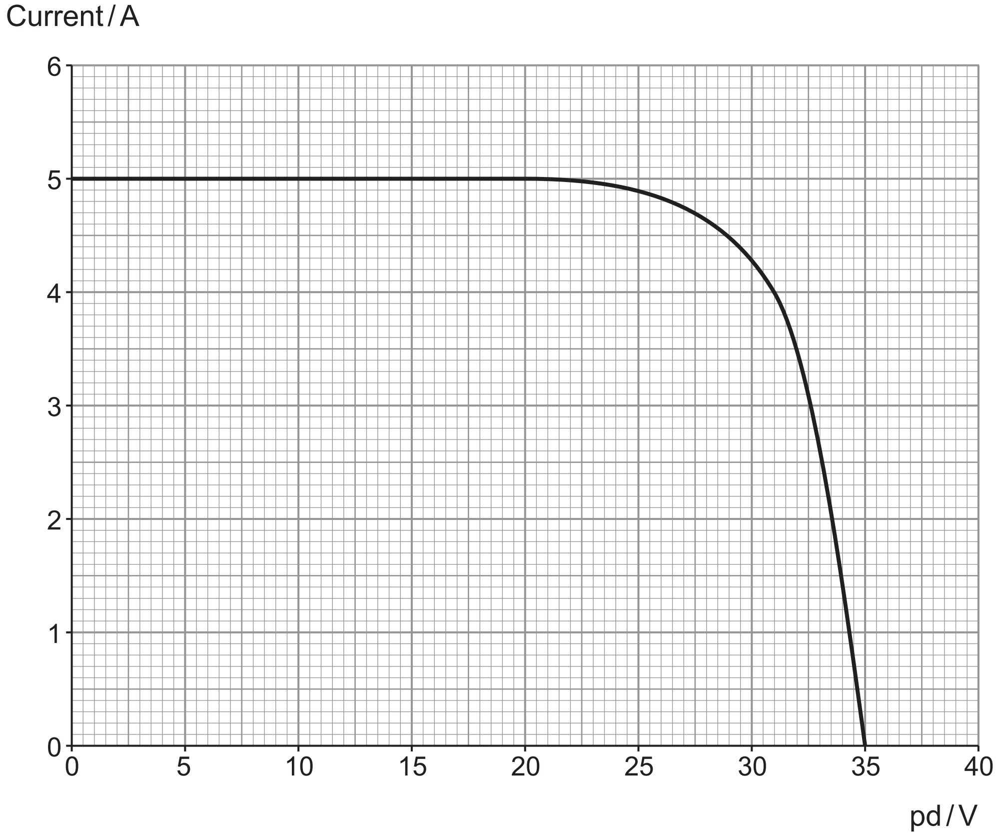
(ii) A solar panel provides electrical power for a load resistor. The graph below shows the current-potential difference characteristics of a solar panel operating under an intensity of 600 W m^{-2}.



The solar panel has a surface area of 1.2 m^2. Determine the efficiency of the solar panel when it is operating at its maximum power output. [4]

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(d) (i) The ground floor of a house has an area of 60 m^2. The floor is made from concrete of thickness 120 mm. If the top surface of the concrete is 19 °C and the lower surface is 5 °C, show that the rate of heat flow through the concrete is approximately 5000 W. K_{concrete} = 0.700 W m^{-1} K^{-1} [2]

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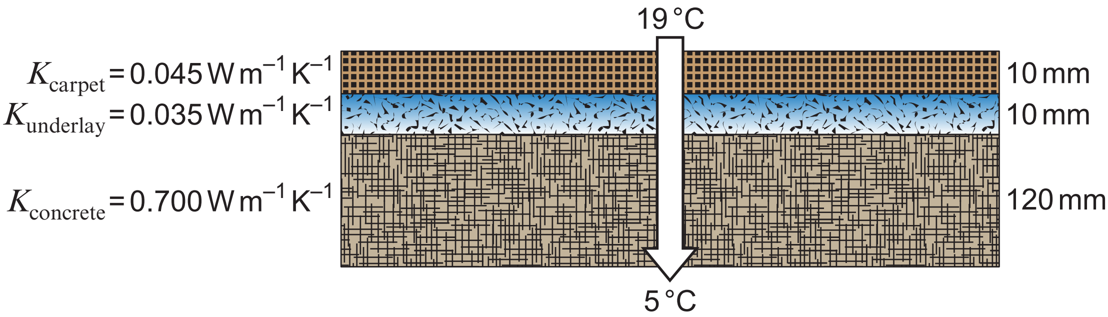
(ii) A carpet fitting company adds underlay and carpet on to the concrete as shown in the diagram.

Diagram not drawn to scale



It is stated by the carpet fitting company that the rate of heat flow through the floor will reduce to a quarter of the value calculated in part (d)(i). Evaluate if this claim is correct. [3]

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